Team 10229

FLORABYTE

Ahmed Anwar

Asser Hatem

Kyrollois Youssef

Yousef Reda

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Chapter 1

Present and Justify a Problem and Solution Requirements

<u>Egypt Grand Challenges</u>

Egypt faces a range of big challenges that need urgent attention to ensure a sustainable future, **as shown in (Figure 1.1)**. The country's Vision 2030 plan aims to guide its development by addressing key economic, societal, and environmental issues. However, rapid population growth and limited progress in technology make it harder to solve these problems. To move forward, Egypt needs innovative solutions that can tackle its challenges while promoting long-term sustainability.

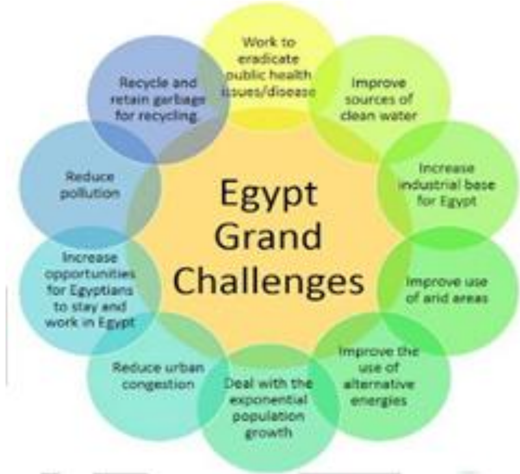


Figure 1.1: The 12 grand challenges that Egypt faces

One major challenge is the fast-growing population, which puts pressure on housing, healthcare, and education. Rapid urbanization is another issue, leading to overcrowded cities and strained infrastructure. Economically, Egypt is working to diversify its economy and reduce its reliance on old industries. Environmental challenges are also significant, with pollution affecting the air, water, and land. At the same time, energy demand is increasing, making a shift toward renewable energy sources essential for the future.

Other challenges include the management of solid waste whereby the current systems cannot support the increased levels of waste being generated. The way such challenges are approached largely dictates the efforts made for securing the future development of Egypt. Now, let's dive into five specific grand challenges out of the eleven that face Egypt: we will take a closer look at their root causes and impacts for ensuring that the nation attains sustainable development and economic balance.

Increase the Industrial and Agricultural Bases of Egypt

Egypt's agricultural sector plays a vital role in sustaining livelihoods, ensuring food security, and contributing to economic development. However, its growth is severely hindered by water scarcity and wastewater mismanagement, two interlinked challenges that threaten both agricultural productivity and environmental stability. With population growth, climatic change, and inefficient irrigation methods placing more pressure on Egypt's water resources, there is an urgent need to adopt sustainable water management strategies. Without these interventions, the country risks declining agricultural yields, worsening rural poverty, and further environmental degradation. One of the fundamental barriers to sustainable agriculture in Egypt is land fragmentation, where farms are divided into smaller and smaller plots due to inheritance practices and population growth. Today, approximately 80% of agricultural landholdings are five feddans or smaller, making it difficult to implement large-scale, modern farming techniques. In many cases, these fragmented lands prevent farmers from adopting advanced irrigation systems, as small plots do not justify the high costs of installing efficient water distribution networks. Instead, farmers rely on traditional surface irrigation methods, which are not only inefficient but also lead to excessive water loss through evaporation, runoff, and deep percolation. Land fragmentation also limits the potential for community-based water management programs. In regions where farmland is divided among numerous smallholders, cooperation is often difficult, leading to uncoordinated water usage, over-extraction of groundwater, and inefficient drainage systems. Furthermore, small-scale farmers lack access to financial resources and technical knowledge that would enable them to implement water-saving technologies. As a result, agricultural water consumption is extremely high, even though freshwater availability is decreasing at an alarming rate.

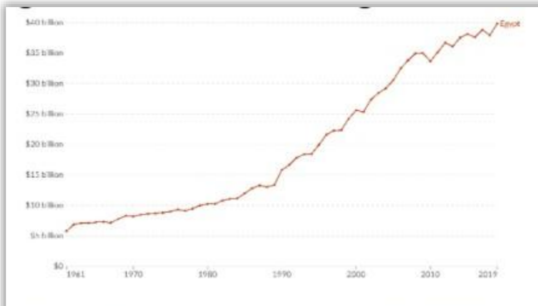


Figure 1:2: Total agriculture output,1961-2019

Causes

Water Scarcity and Outdated Irrigation

Egypt is one of the most water-scarce countries in the world, with per capita freshwater availability far below the water poverty threshold. However, methods of irrigation practiced in Egypt remain largely outdated and wasteful, further worsening the severity of water shortages. The most widespread method of irrigation in Egypt is flood irrigation, where farmers manually flood their fields with large volumes of water. Although this technique is simple and cost-effective in the short term, it leads to massive water wastage, as nearly 50% of the applied water is lost through evaporation or inefficient absorption into the soil.

Groundwater over-extraction further intensifies the crisis. Many farmers rely on unregulated well drilling to access groundwater, often pumping excessive amounts to compensate for surface water shortages. This has led to the depletion of underground aquifers, rising salinity levels, and soil degradation, making large portions of farmland unsuitable for cultivation over time. The continued overuse of chemical fertilizers and pesticides also contributes to groundwater contamination, creating long-term ecological imbalances that affect both crop productivity and human health.

Wastewater Mismanagement

The lack of wastewater treatment facilities in rural areas implies that the majority of farmers have no alternative but to water their crops with polluted water, which contains heavy metals, harmful bacteria, and excessive nutrients. In the long term, this practice degrades soil quality, reduces agricultural yields, and increases the risk of foodborne diseases. Furthermore, excessive agricultural runoff—high in nitrogen and phosphorus from chemical fertilizers—causes eutrophication, a process in which excessive algae growth depletes oxygen levels in water bodies, killing fish and other aquatic life.

Additionally, the problem of sludge disposal from wastewater treatment plants remains unresolved in many regions. Properly treated sludge can be a valuable resource for soil enrichment, but in Egypt, a large portion of it remains untreated and is either dumped into landfills or discharged into water bodies, exacerbating pollution. The inefficient handling of wastewater not only wastes a potentially 19 valuable source of irrigation water but also contributes to further deterioration of the country's already fragile water resources.

Impacts

Reduced Agricultural Productivity

One of the major obstacles to implementing sustainable water management solutions is the economic vulnerability of rural farmers. Many farmers operate on narrow profit margins, making it difficult for them to invest in advanced irrigation systems, water filtration technologies, or wastewater treatment facilities. The high initial costs associated with drip irrigation, hydroponics, or decentralized wastewater treatment units discourage farmers from adopting these solutions, even though they would save water and improve long-term productivity.

The lack of financial incentives, subsidies, or access to low-interest loans further restricts farmers' ability to transition toward sustainable agricultural practices. Government support for smallholder farmers remains limited and often poorly targeted, preventing many from benefiting from technological advancements in water conservation. Additionally, the absence of efficient water pricing mechanisms means that water is often underpriced or freely accessible, leading to widespread overuse and inefficiencies in allocation. Without economic reforms that make water-efficient technologies affordable and accessible, the cycle of water wastage and environmental degradation will continue.

Environmental Degradation

One of the most effective ways to improve water management and wastewater recycling in rural Egypt is through agricultural extension services that educate farmers on modern irrigation techniques, sustainable water use, and wastewater treatment methods. These services play an important role in narrowing the gap between scientific research and on-the- ground agricultural practices, ensuring that farmers have access to the knowledge and resources needed to implement efficient water management strategies.

Economic Vulnerability of Farmers

Training programs should focus on teaching farmers how to integrate wastewater treatment into their agricultural activities, using techniques such as biofiltration, sedimentation, and phytoremediation (using plants to remove contaminants from water). Additionally, the introduction of decentralized wastewater treatment units in rural areas can provide a sustainable alternative water source while reducing pollution levels.

Furthermore, demonstration farms and farmer cooperatives can serve as knowledge-sharing platforms, where farmers can acquire skills to install precision irrigation, water recycling schemes, and rainwater harvesting. Strengthening the capacity of extension workers by equipping them with technical expertise and 20 communication skills will ensure that farmers receive tailored advice on water-efficient farming methods.

Recycle Garbage and Waste for Economic and Environmental Purposes

Recycling garbage and waste, at its core, is a process of collecting and processing materials that would otherwise be thrown away as trash and turning them into new products. Moreover, there are many justifications for recycling garbage and waste, including environmental protection, economic advantages, and fighting climate change.

Egypt occupies the 7th position in the list of countries with the most mismanaged plastic waste ***(Simpson*, 2024)**. Moreover, Egypt could be classified as one of the most countries that produces plastic waste, **as shown in Figure (1.3)**.

Figure 1.3 represents the ratio of plastic mismanagement in each country in the world.

Egypt recycles only about 6% of its waste, while the average global recycling rate is estimated at 19%, and the African average is 11% ***(Okafor*, 2024)**. Additionally, there are about 28 recycling factories in Egypt. This number is a little small in comparison with the African average of factories, which is estimated at 37 factories ***(Embaby*, 2023)**. These could cause issues, as follows:

Firstly, plastic waste production is estimated at 3 million tons from 2018 to 2021***(AfricaNews, 2023)***. While 43% of this waste production is disposed of by throwing it into the Nile River, the Red Sea, and the Mediterranean Sea. That's why the number of people dying due to drinking from this water has risen to 19,200 people ***(WorldBank*, 2019).**

Secondly, the lack of recycling of garbage and waste contributes to Egypt being ranked 12th in air pollution. Additionally, the PM2.5 level is about 39.77 µg/m³. Furthermore, the air quality index (AQI+) is estimated at 112 AQI+ in the US. That's why Egypt's air quality is classified as moderate. ***(Egypt Air Quality Index (AQI) and Air Pollution Information | IQAir*, 2020).**

Thirdly, plastic waste could be considered a new stressor on soil, in the form of microplastics (MP_S). A percentage of microplastics (MP_S)can affect the cycling of carbon compounds and other biogenic elements by impacting the soil microbiome, enzyme activity, and plant growth. Consequently, this can increase the emissions of CO_2 and CH_4**, in Figure (1.4), *(Vainberg et al.*, 2025).**

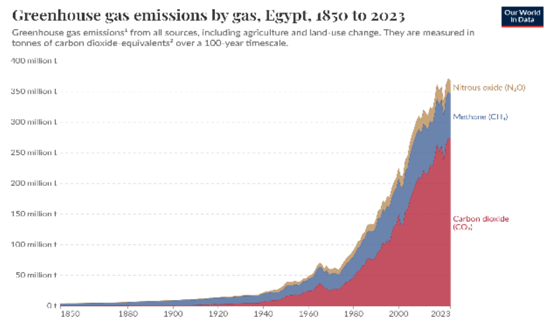


Figure 1. 4: Greenhouse gas ratios

Thus, Egypt suffers from a severe lack of recycling facilities and occupies a high position in the amount of waste produced. These lead to crucial issues, including air, water, and soil pollution, human health issues, and climate change.

<u>Causes</u>

<u>Lack of recycling facilities:</u>

Egypt suffers from an observed lack of recycling facilities, where there are only about 28 factories. This number can be considered a small number in comparison with the average African factory, which is 37 factories. Consequently, this enabled Egypt to recycle only 6% of its waste, which is estimated at 180,000 tons ***(Okafor*, 2024)**. Additionally, the lack of recycling significantly increases the cost associated with raw material procurement. For instance, the lack of recycling material, like aluminum, which is only about 6% recycled, could lead to an enormous expansion in energy **(*Recycling: Recycling as a Source of Raw Materials: Closing the Loop in Manufacturing - FasterCapital*, 2025).** Hence, governance will incur additional costs in manufacturing these products.

<u>Absence of Modern Technologies and High Initial Costs:</u>

Not all materials can be recycled infinitely, and some require more energy and resources to recycle than others. Additionally, nearly all recycling facilities require a significant initial investment.

First, plastic recycling demands advanced technology to handle different types of plastics. Moreover, the equipment needed, such as shredders, pelletizers, conveyors, mixers, and other technology needed for plastic recycling, lies in the range of 30,000 to over $300,000 (Eco Recycling Today, 2024).

Second, mixed materials, which may include both recycled and non-recycled materials. Hence, this requires advanced technology to separate them and recycle them ***(Directory*, 2025*)*.**

That's why the lack of modern recycling technologies is considered the main reason for the low percentage of recycled material in Egypt, which is only about 6%. Additionally, high initial costs are a major factor behind the scarcity of recycling factories in Egypt, with only 28 factories compared to the African average of 37 factories ***(Embaby,* 2023*)*.**

Educational Issues :

Lack of education, which is known as illiteracy. The high percentage, approximately 75%, of illiterate individuals, who struggle to operate the modern technology used in recycling factories, which approaches 74.5%, **as shown in Figure (1.5),** is a significant factor contributing to the low recycled material percentage, approximately 6%. This contributes to making Egypt suffer from serious issues, including increased air pollution due to the burning of waste, deterioration of marine ecosystems through plastic contamination, escalating climate change as landfills release harmful greenhouse gases, and leading to human health issues. Additionally, an increase in plastic waste can contaminate the soil in agricultural areas by absorbing microplastics (MP_S). Consequently, plants would be prevented from properly absorbing the nutrients, thereby impeding their growth ***(De Boer,* 2023*)***. Thus, this percentage of illiteracy leads to serious crises.

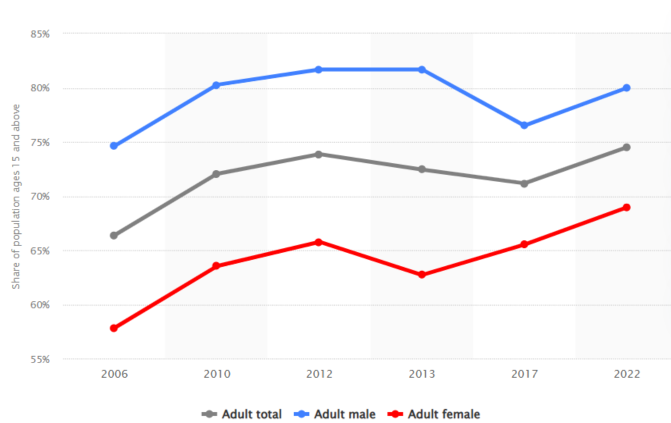


Figure 1.5: The ratio of illiteracy according to gender in Egypt.

Impacts

Air Pollution:

Air pollution is the contamination of the indoor or outdoor environment that could be an impact on various things, such as the lack of waste and plastic recycling. Air quality in Egypt is classified as moderate due to the high percentage of air quality index (AQI+), which is estimated at 112 AQI+ US. Moreover, the high concentrations of PM 2.5 rose to 39.77 µm/m^3, which is eight times the average global concentration, **as shown in Figure (1.6)**. Consequently, Egypt occupies the 12th of air pollution ***(Egypt Air Quality Index (AQI) and Air Pollution Information | IQAir*, 2020)**. That's why if a person inhales an area full of PM 2.5 and PM 10, a person could suffer from diseases like asthma and stroke.

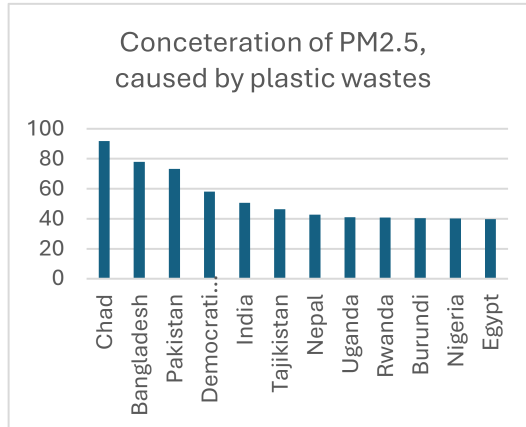


Figure 1.6: Concentration of PM2.5 across several countries.

Marine life damaging:

Lack and limitation of recycling facilities can be considered one of the main factors in the following crises:

Firstly, Egypt produces about 3 million tons of waste, while 43% of this waste production is disposed of by throwing it into the Nile River, the Red Sea, and the Mediterranean Sea. That's why water bodies are in a dire state due to this massive strain from pollution. That's why the fish production in Egypt decreased from 2,038.9 to 2,0.16 metric tons, **in Figure (1.7), (Statista, 2025)**. Additionally, reefs are shrinking hard, whereas coral reefs cover declined by 13.6% on average between 2005 and 2019 **(*The State of Egypt&Rsquo;S Coral Reefs | Enterprise*, 2022)**.

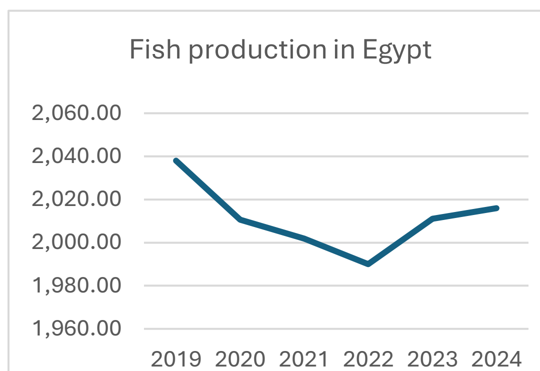


Figure 1.7: Fish production in Egypt, 2019-2024

Next, marine plastic pollution has increased 10x since 1980, affecting at least 267 animal species, including 86% of marine turtles, 44% of seabirds. Also, this pollution affects about 43% of marine mammals **(*Plastics and Biodiversity | Plastics and the Environment Series*, 2025).**

Climate change:

Climate change refers to long-term shifts in temperatures and weather patterns. Climate change could be impacted by the lack of plastic recycling, the percentage of microplastics, and an increase in greenhouse gas emissions. The percentage of plastic and garbage recycling is estimated at 6%, which is below the global average, leading to an increase in the percentage of microplastics in soil and air. Consequently, soil releases more CO2 and CH4 emissions, **as shown in Figure (1.7), (Vainberg et al., 2025)**. This has caused a serious increase in carbon dioxide concentration from 375.924 to 412.727 ppm, **as illustrated in the Figure(1.8)**. This increase in concentration can lead to several diseases, including blood acidosis and acute respiratory acidosis. Hence, the boredom of recycling facilities leads to serious issues. Additionally, the rate of global temperature increase has risen from 0.06°C to 1.5°C over the last century **(Climate Change: Global Temperature, 2025).** Furthermore, about 1,000 people are expected to die due to climate change by 2050.

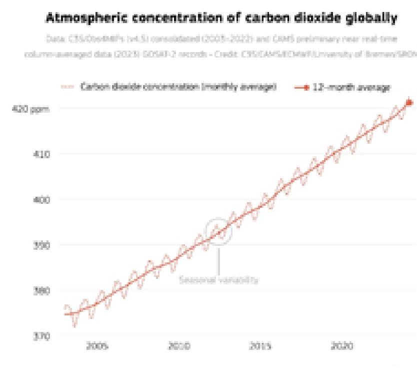


Figure 1.7: Atmospheric concentration of carbon dioxide.

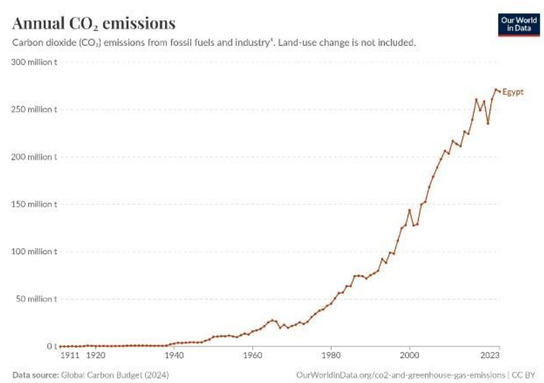


Figure 1.8: Annual CO_2 emissions

Deal up with Population Growth

Population growth refers to an increase in the number of people in a population, which happens because the result of the birth rate being higher than the death rate. The world passed 8.1 billion people in 2024, and Egypt has a population of about 106 As shown in **Figure (1.9),** Egypt's population increased from 26.63 million to 106.9 million from 1960 to 2024.

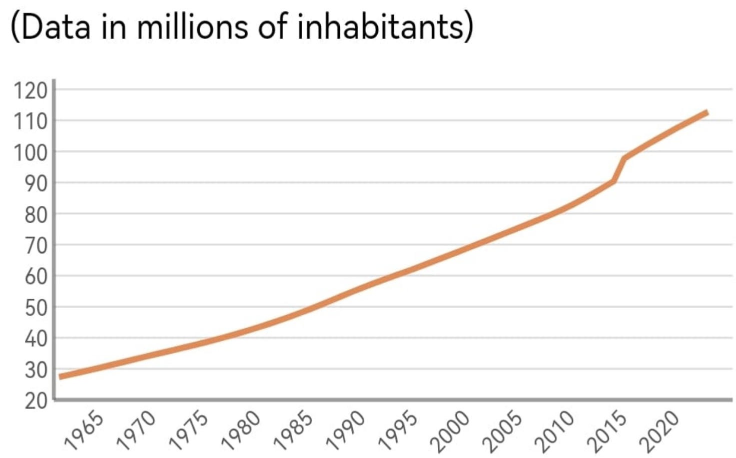


Figure 1.9: Explain the increase in population in Egypt over the last 55 years

This is a growth of 323.2% in 63 years**.** The highest increase in Egypt was recorded in 2015, with 8.07%. Egypt's fertility rate is about 3.097 births per woman, so Egypt has a high percentage of population growth. In the same period, the total population of all countries worldwide increased by 164.7%, well above the global population growth rate of 2.1 births per woman. The percentage of population growth in Egypt is much higher than the percentage of population growth worldwide. The most prominent Egyptian governorates with population growth are Cairo, Giza, and Sharqiyah. The population in Cairo is about 10.3 million people, the population in Giza is about 9.6 million people, and the population in Sharqiyah is about 8 million people. The high population in regions and other regions can have serious impacts and result in an increase in the cost of building facilities.

Causes

Customs And Traditions:

One of the reasons for the population increase is the increase in the number of births with improved medical conditions, in addition to the customs and traditions that support early marriage, the preference for male offspring, direct childbearing in marriage, the lack of use of family planning methods, and the lack of only two children. This lack happens when the males and females suffer from decreased

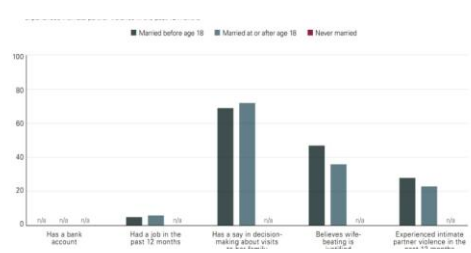


Figure 1.10: Ratio of early marriage

education. The culture of childbearing that has been established in Egyptian society for a long time is still linked to a lack of power and support.

Egypt and African countries have a long history of early marriage, there are males and females under the minimum age (18 years) for each other, they get married, and this is one of the severe causes of population growth, high health risks, and illiteracy. In Egypt, the census shows that over 111,000 women and girls were married before the age of 18, **Figure (1.10)**, with 84 percent coming from or living in rural areas.

Illiteracy:

Lack of education, which is known as illiteracy, is very dangerous to the community because illiteracy is one of the causes that leads to population growth. The illiteracy rate in Egypt stands at 24.7% reflecting the percentage of the adult population aged 15 years and above. In Egypt, the illiterate people are estimated at 14.3 million people, as shown in **Figure (1.11).**

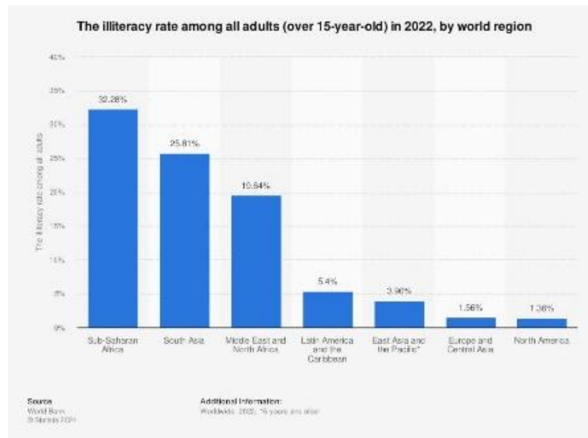


Figure1.11: The ratio of illiteracy

Data from the Egyptian Family Health Survey 2021 that the average optimum number of children among the majority of minors who participated in the survey sample who had never been married is still high, reaching 2.6 children. The optimum number of children among young men is about 2.7 children and the optimum number of children among women is about 2.4, this is less than the men about 0.3 children.

Fertility Rates and Mortality Rate:

The fertility rate is the average number of babies born to females during their reproductive years (generally considered 15 to 44 years), the most commonly used metric is the total fertility rate (TFR), which measures the average number of children per woman, the total fertility rate in Egypt was around 3.097 children per women as shown in **Figure (1.12)** when the global average fertility rate is around 2.3 children per women.

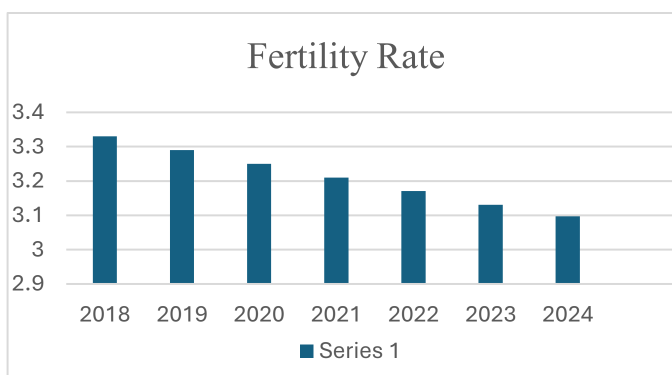


Figure 1.12: The fertility rate

Mortality rate is a measure of the number of deaths in a particular population scaled to population per unit of time, the death rate in Egypt is estimated at 6.39 deaths per 1000 inhabitants while the global rate is estimated to be 8.72 deaths per 1000 inhabitants, due to the decrease in mortality rates and the increase in the fertility rate in Egypt, Egypt suffers from population growth and increase in the number of people.

Impacts

Cost of Health care and Education facilities:

The latest statistics from the Information and Decision Support Center of the Council of Ministers indicate that the total value of public spending on the health sector in the budget for the fiscal year 2021/2022 reached about 108.8 billion EGP. The budget increased rate of 15.3 billion EGP (about 16%) over the allocations of the health sector in the fiscal year 2020/2021, which amounted to about EGP 93.5, and in 2023/2024 spending reached EGP 75 billion. This high cost occurs because the population growth makes Egypt need more hospitals and other health facilities.

Egypt had population growth with a dominating younger generation. This would generally entail increased pressure on the education system in Egypt, which would lead to building new schools for the new generation, and the cost of marketing schools from 2020 to 2024 will increase from 2.107 billion to 4.943 billion, as expected.

More Burning of Fossil Fuels and Environmental Impacts:

The relationship between population and the environment is a significant issue due to its impact on the chances for achieving sustainable development, especially in developing countries. Previous studies on relationships have primarily focused on the impact of population growth on the environment. In Egypt during 1950 / 2020, the current study finds that in Egypt, firstly, 1% increase in population raises the CO_2 emissions by 2.4%, and secondly, an increase in CO_2 emissions by 1% is associated with an increase in our air pollution emissions.

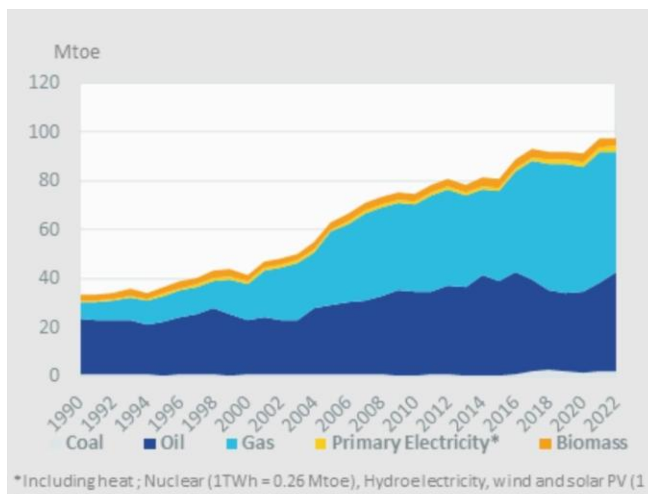


Figure 1.13: The amount of burning fossil fuels

This increase in pollution happened due to population growth which makes Egypt's government need to use more fossil fouls for energy and people need more transport and generate electricity, which needs more petrol and oil. These materials have many harms and impacts like an increase in local pollution and spreading disease. **Figure (1.13)** shows the amount of fuel the country burns.

Economic Strain:

Strain theory, in sociology, proposes that pressure is derived from social factors, such as lack of income or lack of quality of facilities. In Egypt, the economy has been growing since the habitats need more materials such as metal, petrol, and other materials which have a big important in life. These needs force Egypt to import these materials from other countries, which has negative consequences on the Egyptian economy. In Egypt, the real annual economic growth rate reached approximately 3.8% in the previous fiscal year (2022/2023) and the economic growth in Egypt decreased from 9.8% in 2021/2022 to 4.4% in 2022/2023. The economic growth in 2024 is estimated at 2.22%, and this is less than the limit according to the central bank, as shown in **Figure (1.14)**. This strain happens because people will need more materials, and these materials will be more than the available source materials in Egypt.

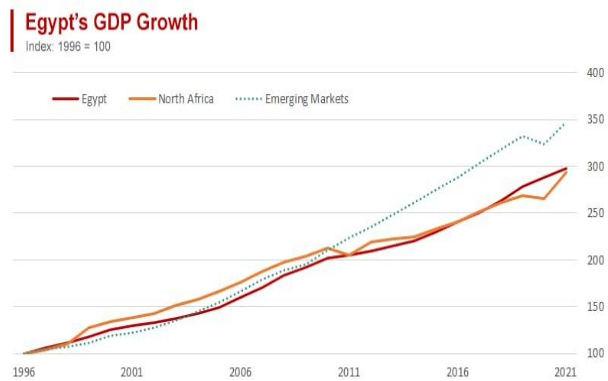


Figure 1.14: Egypt's growth economy

Address & Reduce Pollution in Air, Water, and Soil

The importance of pollution control and waste management in Egypt has become increasingly important. Pollution happens in many forms, such as air, water, and soil pollution. These are the most common forms in which pollution occurs. Pollution has control in these because greenhouse gases are increasing these days. Greenhouse gases have become very important because climate change has become an overrated variable these days.

Pollution is a global issue caused by many factors, such as weak environmental laws and controls, government neglect, and lack of awareness. Egypt contains a large amount of pollution because of some factors such as vehicle emissions, poor waste management, industrial pollution, and many other things.

Also, water pollution is an important issue, especially in the Nile River that considered the main water source in Egypt. Another form of pollution is soil, and it happens because of the chemical compounds that are put in agricultural lands. Pollution will continue to be dangerous for public health, the environment, and the economy if we do not have effective pollution management.

Causes

Industrial pollution

Multiple factors contribute to the exacerbation of air pollution in Egypt. One of the major causes of environmental pollution is industrial pollution activity in many governments, such as Cairo and Alex. Factories release large amounts of toxic gases such as sulfur dioxide, carbon monoxide, and nitrogen oxides into the air or atmosphere, which leads to air pollution and increases diseases and damage. Also, the rise in traffic and infrastructure further increases air pollution levels, which leads to the death of many people in Egypt, as shown in **the Figure (1.15).** However, agricultural habits, like burning crops in the air, also contribute to chemical pollutants in the air.

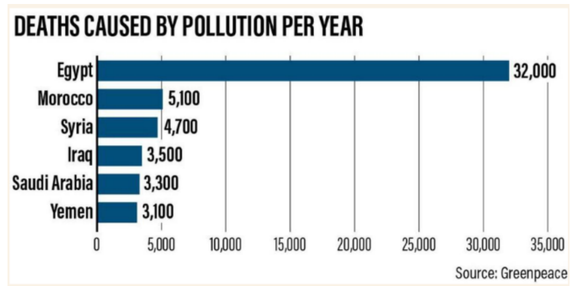


Figure1.12: The deaths caused by pollution per year

Additionally, some factories discharge polluted water directly into rivers and seas. This water contains heavy metals, chemicals, and plastics, which make the water not suitable for drinking. According to the World Bank, industrial emissions contribute up to **30%** of air pollution in urban areas of Egypt. Putting this into perspective, regulations and efforts need to be put in place to stop the bleeding.

poor waste management

Improper waste disposal is a significant cause of environmental pollution, affecting air, water, and soil. In many places, industrial and household are disposed of incorrectly, such as being dumped in open landfills or water bodies without proper treatment. This leads to the accumulation of plastic, toxic chemicals, and organic waste in Egypt, as shown in **the Figure (1.16).**

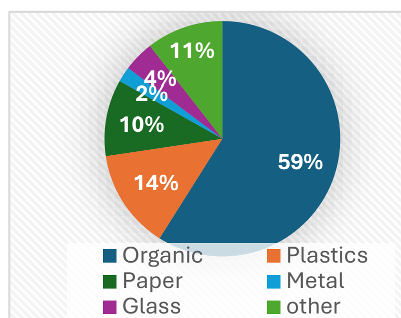


Figure 1.13: Composition of municipal solid waste in Egypt

Biodegradable wastes constitute **56%,** followed by inserts **15%**. Most of the developing countries, like Egypt, generate more biodegradable waste compared to other types of waste, and this leads to soil pollution**.**

One of the dangerous effects is soil pollution, as harmful substances from waste seep into the ground, which reduces its fertility and affects plant growth. Also, incorrectly managed waste leads to the blocking of water channels which leads to water pollution and the spread of diseases. And these things spread in Egypt because of a lack of awareness.

Vehicle Emissions

Vehicle emissions are the most common thing that can cause air pollution, which releases harmful gases into the atmosphere. Vehicles are a main source of transportation greenhouse gas emissions from vehicles is nowadays increasing. Vehicles produce gases such as carbon monoxide, nitrogen, and oxides because they burn fossil fuels which contribute to producing fog, and acid rain.

In some cities, like Cairo, traffic leads to increased fuel use and releases harmful gases, which put people in danger because it is crowded. Additionally, some old cars or not fixed correctly, leading to the release of large amounts of pollution because of burning fossil fuels. In some countries like Egypt, the number of cars is increasing by 12.02% during the first 10 months of 2024, with total sales reaching approximately 78,400 vehicles, which increases pollution.

Impact

climate change acceleration

Climate change is the result of pollution, it happens because of large amounts of releases of gases such as carbon dioxide, methane, and nitrous oxide. These gases are dangerous of humans and animals because they are leading greenhouse gases and a rise in the level of temperature. For example, the burning of fossil fuels, like coal, and petroleum. Climate change happens by increasing gases because of pollution as shown in **Figure (1.17).**

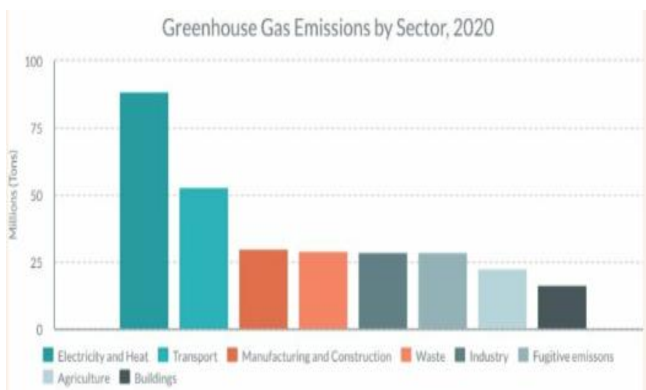


Figure1.14: Gas emission in Egypt.

When pollution affects climate change, weather conditions become hard such as burning, volcanoes, and floods which cause environmental damage. Also, climate change affects on melting of ice at the poles which leads to rises in sea level and damage to cities such as Alex and Hurghada.

Health Problem

Pollution has an impact on human health, especially due to air, water, and soil. Health problems that impact pollution are very dangerous because they effect on respiratory system, digestive system, and public health care of humans. Environmental pollution like air pollution and water pollution poses a big challenge in Egypt because they effect on health care of humans.

Air pollution occurs due to many things such as emissions from vehicles, and the burning of fossil fuels which lead to psychological diseases and damage. It was the cause of **90,559** premature deaths in Egypt in 2019 and more than **12%** of all deaths in 2017. Water pollution results in the throwing of waste which harms marine creatures and makes water not suitable for drinking. Soil pollution happens because of many things, the most important are waste and pesticides.

Environmental Degradation

Environmental degradation is the decay of the ecosystem caused by human consumption of resources such as air, water, and land. The most effective factor is climate change which is caused by some factors such as weather fluctuations, and ecological pollution. Environmental degradation consists of both pollution and the degradation of resources. Our activities have an impact on the environment. So environmental degradation should be considered a major problem. Egypt faces issues from air pollution, river pollution, garbage, and environmental pollution.

Economic

Pollution does not only harm the environment and human health, but it also causes economic losses for the government. These losses are shaped by some things such as increased healthcare costs, low productivity, and tourism damage or decline. Firstly, healthcare costs due to treatment and hospitals because pollution leads to the spread of diseases in the respiratory system. And this makes the government save money to face these diseases and it reduces the resources. Secondly, declining tourism is caused by polluted places like beaches that suffer from water pollution. So, pollution makes reduces the attraction of tourists. Also, soil pollution is caused by adding chemical components and insecticides leading to reduced quality of crops. Additionally, water pollution dead fish and sea animals which reduces fish and effects on economics of fishers

Table 1 :ESTIMATED ANNUAL COST OF HEALTH EFFECTS OF INADEQUATE WATER, SANITATION, AND HYGIENE IN EGYPT,

	Low	Central	High
Cost of mortality	13.3	20.2	28.1
Cost of morbidity	12.8	19.2	27.9
Total cost of health effects	26.1	39.4	56.0
% equivalent of GDP, 2016/17	**0.75%**	**1.14%**	**1.61%**

According to the World Bank, the annual cost of the health effects associated with inadequate drinking water, sanitation, and hygiene is estimated at **LE 26 billion** to **56 billion** in 2016/17, with a central estimate of LE **39** billion **as shown in table**. The cost is equivalent to about **0.75%** to **1.61%** of Egypt's GDP that year, with a central estimate of **1.14%.** Recent studies estimate the annual economic cost of air pollution to health in the Greater Cairo Region alone **at 1.4 per cent** of Egypt's GDP.

Problem to Be Solved

Limited Water Availability and Low Soil Quality for Sustainable Agriculture in 6th October City

The severe lack of plastic waste recycling is a core problem in Egypt. According to recent data, Egypt occupies 7th position in the list of countries with the most mismanaged plastic waste ***(Simpson,* 2024)**. ~~As a direct result, the soil and water contamination levels will increase, and the temperature as well.~~ Consequently, this waste accumulation directly exacerbates the contamination of soil and water, and arid areas. Evidently, the temperature reaches 49.0 °C; due to the increase in greenhouse gas emissions, especially CO_2 **(WorldData.info, 2026).**

Thus, this led to altering the precipitation patterns and shifting the ecosystem dynamics in Egyptian arid areas **(Greenfield & Greenfield, 2024).** On the other hand, the average required temperature for many crops in Egypt is 20-28 °C.

Narrowing the scope to the Giza Government—especially 6th October City—we found a city has an extensive desert climate. Clearly, its temperature averages 36.0 °C, while the average precipitation is approximately 0.1 mm/day. In contrast, the average temperature and precipitation needed are 30 °C and 7.5 mm of water per day, respectively.

The city of 6th October is a city with high population density, which is estimated at 3,112 people/km on about 53,744 acres **(*6th of October City » EBRD Green Cities*, n.d.)**. This huge population consumes about 150-200 ft^2 per capita **(Lisa, 2025)**. At the same time, the city of 6th October produces a small percentage of its crop production.

Furthermore, the city of 6th October lies far from the main water source (the Nile River), with a distance estimated at 20.5 km. This distance led to essential consequences, as follows:

Firstly, the low rainfall and the city's location away from the Nile River caused desert conditions. Over time, soil moisture decreased by 1%; meanwhile, the moisture in agricultural lands ranged from 2% to 30% **(Mohamed et al., 2019).**

Secondly, the low water quality as the city depends on groundwater. Moreover, transporting Nile water over a distance of 20.5 km introduces risks of contamination, which poses maintenance issues in water quality.

Thus, the problem that should be addressed is how to maintain a high-quality environment for agricultural fields in the city of 6^{th} October. This is achieved through optimizing the hydrogeological and climate conditions, which currently prevent sustainable agriculture production.

Negative Consequences

Public health & pollution:

Egypt has negative results from climate change. One of the main benefits of smart greenhouse is that it can reduce pollution and improve air quality by using recycled materials for building the prototype such as plastic for structure and some of electronic waste. Environmental pollution like air pollution and water pollution poses a big challenge in Egypt because they effect on health care of humans but by using smart greenhouse recycled project, the air and water will become clean and not polluted. On the other hand, in addition to reducing air pollution, repurposing old plastic and salvaging electronic components is very important.

Lack of Recycled Issues:

Scientific environments like not-recycled greenhouse require a lot of technological machines to provide farmers with the opportunity to conduct scientific experiments and analyse the results for a better quality of production. These smart greenhouse projects are significantly above the country and farmer's budget. For instance, the greenhouse requires some equipment, such as sensors and expensive structure, which provides the chance for innovative farmers and governments to apply their creative designs, projects, and prototypes to the real world. This is an example of only structure of the smart greenhouse not recycled. As same as smart greenhouse, other smart agriculture, also need some technological devices, which is used in maintain production. Similarly, these are high-cost devices, highlighting the same issue.

Seasonal Limitation of Crop Production:

Traditional agriculture in Egypt is depending on seasons, because the farmers rely on the natural climate change such as seasonal temperature, humidity, and sunlight. The farmers can only grow crops during certain time of the year. This seasonal limitation prevents the availability of production in all year which make it high demand or sensitive crops, like strawberries or specific flowers. Without automated closed-loop smart greenhouse systems to maintain environmental conditions like change temperature. Egypt's agricultural output remains heavily restricted by the calendar, making it difficult to increase the agricultural base and secure year-round food availability.

Positive Consequences

Cost Efficiency:

The greenhouse project provides a way to design and build a highly effective solution at fraction of the cost of commercial systems. This project attracts more international and local investment, which is estimated in the millions. Greenhouse projects save money for Egypt because they are made from recycled materials, and agriculture instruments are often expensive and difficult to maintain. Also, it is useful for citizens to start a new job. Greenhouse project considers generating jobs like separating electronic components from recycled materials, project development, manufacturing, and sales. It offers opportunities to workers possessing various skills, including engineers, electricians, and other workers. Additionally, when the project becomes available at a low cost and small scale, farmer students can learn how it works and technicians to build.

Reducing Waste:

This project helps reduce pollution and waste by reusing old plastic and components to build new projects. Every year, millions of electric and electronic items are discarded. One of the main benefits of recycling is that it can reduce pollution because it does not release any emissions into the air. Using recycled materials in this project will save money for the government, instead of being thrown away or burned release toxic substances. Recycling old things protects public health by reusing and repurposing components like motors, sensors, and circuit boards. Repurposing old plastics for the structure and salvaging electronic components will minimize e-waste generation and prevents these hazardous materials from entering the environment.

Enhancing Agriculture:

The implementation of an automated closed-loop greenhouse control system allows for monitoring of parameters, such as soil moisture, humidity, temperature, and light. By utilizing real-time measurement feedback, the system triggers immediate corrective actions through specific actuators. This adaptive mechanism effectively prevents sudden environmental fluctuations, thereby restoring and maintaining the optimal conditions required for sensitive and productive crops. By build this project with optimal environmental conditions to support agricultural production. This system will make agriculture food available in all seasons which increase the economics of Egypt.

Educational:

The small-scale smart greenhouse project has many benefits for education, especially in schools and university that depend on projects. Building a greenhouse in Egypt will help students to apply and gain experience in many areas, such as electronics, programming, agriculture, and design. The greenhouse is a future project that helps farmers and students to make smart farm or experiments. Also, the smart greenhouse will allow students to understand its mechanism and use it with accuracy and safety. This fosters a generation capable of designing low-cost, sustainable tech solutions for the future.

Research

Topic Related to the Problem

Climate Change & Greenhouse Gas Emissions in Egypt:

Climate change is a major problem facing the world. The cause of it is due to greenhouse gas emissions. Egypt contributes about **0.63–0.75% of global emissions** (of about 440 million tons CO_2eq in 2022). And these percentages increase from the earlier baselines by more than 44%. Consequently to those emitted gases, the temperature in EGYPT increases by 0.38 °C per decade, leading to water scarcity from evaporation, drought, and saltwater intrusion. So, this acts as a main role in the reduction of about 30% in food production by 2030. The main factors of the emissions of greenhouse gases are plastics contributing about 10-14% of Unrecycled municipal solid waste (MSW), which leads to CO_2 and trace methane release during incineration or long-term burial. Moreover, poor recycling leads to Soil contamination, water contamination, and blocked drainage systems.

Limited Water Availability in 6th October City

One of the main problems which EGYPT faces is the water scarcity, at which the pointer to the water scarcity should be <1,000 m^3 per person per year, and the specialized amount of water per person is about 500 m^3, and it is actually a pointer for the scarcity of water in EGYPT. Moreover, 6th October City's growing population and urban development are competing for extremely limited water resources. There are some drivers for the water strain, through which more population growth, accompanied by urbanization and more buildings, leads to more water demand. In addition, the desert heat increases water loss, leading to less rainfall that then reduces replenishment. Also, the inefficient water usage, which is represented by old irrigation, leaks, or overuse in cities and farms, results in making scarcity worse.

Desert Climate & Poor Soil Quality in 6th of October City:

The desert climate is one of the main problems facing 6th of October City. The city has very hot weather and dry conditions most of the year. The temperature averages about 36 °C and sometimes reaches 49 °C, while many crops need a moderate temperature between 20–28 °C to grow properly. Also, the source of agriculture from the rainfall is extremely low, about 0.1 mm per day. Also, due to the high temperature in the region of Sixth October City, water evaporates quickly, and the soil becomes extremely dry. Moreover, due to the sandy and existence of low organic materials in the soil of sixths october, it cannot hold water or nutrients for a long time. In addition, the city depends mainly on groundwater, which is very remote and located about 20.5 km away from the Nile, which makes irrigation more difficult and limits sustainable agricultural production.

Topic Related to the Solution

Feedback Control Systems in Smart Agriculture:

Feedback control systems are engineering systems that track the outputs of a system and adjust the inputs to ensure that the system remains stable around certain setpoints. In greenhouse farming, sensors measure environmental factors such as soil moisture, humidity, and light intensity. The microcontroller interprets the data and turns on actuators based on rule-based IF-THEN logic with hysteresis. Feedback control systems are closed-loop systems that react to environmental changes, unlike open-loop systems. They prevent conditions such as overwatering, high humidity, and light stress. Feedback control systems in agriculture are directly related to sustainable agriculture and precision agriculture.

Sustainable Construction Using Recycled Materials

Sustainability is one of the requirements of the capstone project. The greenhouse building is made from recycled or reclaimed materials such as wood and plastic sheets. The use of recycled materials has a positive impact on the environment and is part of waste management strategies. The use of recycled materials is cost-effective and helps to solve the environmental problems of garbage accumulation and pollution in the country. In terms of structural integrity, the use of recycled materials is portable and durable enough for prototype development.

Smart Irrigation and Resource Efficiency

Water conservation through efficient irrigation is a critical need in arid zones like Egypt. Smart irrigation technology uses real-time soil moisture analysis to determine when and how much water should be supplied. A drip irrigation network coupled with a controlled water pump ensures efficient watering of plants by pumping water directly to their roots, thus reducing evaporation and runoff losses. The combination of automated irrigation and environmental monitoring ensures optimal water usage while promoting healthy plant growth. The measurement of actuator power consumption enables the assessment of energy efficiency, thus ensuring that the system is environmentally and financially sustainable.

Prior Solutions

Atlantic Sapphire's "Bluehouse"

Overview:

The Atlantic Sapphire's Bluehouse aquaculture, as demonstrated **in Figure (1.18),** is based on a system that aims for sustainable seafood production. This approach is called a recirculation aquaculture technique. This Bluehouse facility was founded in Florida, USA, 16 years ago, with an area exceeding 380,000 ft^2. The cost of materials used throughout the construction is 130 million dollars.

Figure 1.18: The Atlantic Sapphire's Bluehouse Aquaculture

It utilizes 36 massive concrete recirculation tanks—grow-out tanks **(PENETRON, 2026).** Each tank has a capacity of 454,000 gallons. They are chilled to about 59 °F (16 °C). Consequently, the Atlantic Sapphire's Bluehouse is capable of producing approximately 9,500 tons of Salmon annually, roughly 0.34% of the global market **(Annual Report, 2022).** The project has a future goal of reaching about 7.8% of the global market by 2031. Furthermore, the aquaculture consumes about 8.8-10 kWh/kg smolt with an average cost of $0.12 per kWh **(*This Is How Much Electricity Atlantic Sapphire Used per Kilo of Salmon*, 2021).**

Of all the water used, under 5% is freshwater, and over 95% is saline water; this significantly mitigates the consumption of freshwater via replacing it with treated saline water **(Bluehouse Salmon, 2025).** The Atlantic Sapphire's Bluehouse was implemented to provide fish with the optimal environment possible. It is composed of several stages: degassing, oxygenation, disinfection, nitrification, and temperature control. These stages are able to remove the physical and chemical contaminations **(Recirculating**

Aquaculture Salmon Network, 2025). This water is recycled every half-hour and removed every 10 days. To ensure a precise amount of oxygen that must be included in water.

Then, the water rises into the tank via stainless steel pumps. These pumps consume nearly 150 kW—this indicate its high efficiency, since they move up 2,700 liters per second. Ultimately, the Atlantic Sapphire's Bluehouse utilizes advanced artificial intelligence models to optimize the feeding and maximize production.

<u>Mechanism:</u>

The 36 massive tanks utilized in Atlantic Sapphire, **as literature in Figure(1.19)**, each have a capacity of 454,000 gallons **(PENETRON, 2026).** They are filled with water via high-powered stainless-steel pumps of a model MPFR-I chopper pump **(*Atlantic Sapphire Silage Plant With BioChop System | Landia*, 2026)**. These pumps can move approximately 2,700 liters per second through outlet openings in the range between 150 mm and 900 mm.

Figure 1.19: The Atlantic Sapphire's Blue house mechanism

The water that is supplied for each tank is divided into under 5% is freshwater and over 95% is saline water **(Bluehouse Salmon, 2025).** Across aquaculture operations, the water is recycled every half-hour—the recirculation aquaculture technique. **According to Figure 1.20**, there is a confining zone that takes in at 1,000-2,000 ft, which acts as a natural seal. This prevents fresh water from mixing with other layers.

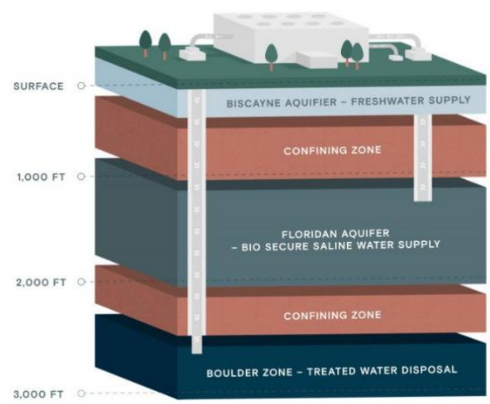


Figure 1.20: The infrastructure of the Atlantic Sapphire's Bluehouse water supply.

Likewise, Atlantic Sapphire depends on saline water (95% of its water supply) from the Floridan Aquifer. After the water reaches its tank, it takes just half an hour to goes upon a treatment process as follows:

There are approximately 50 kg/m^3 of fish inside each tank, which produce gases including: CO_2, N_2, and other gases. In the process of degassing, the CO_2 is removed since the excessive accumulation of it lowers the water pH level **(*Atlantic Sapphire Silage Plant With BioChop System | Landia*, 2025)**.

Then, the oxygen level (O_2) is optimized by injecting pure O_2 liquid. After that, it is vaporized and directed into the system via a narrow pipe, according to the following formula:

$$C = k_H \cdot P \quad \textbf{(1)}$$

Where c is the solubility in mol/L, k_H is Henry's law constant, and P is the partial pressure.

Next, the disinfection process begins; some substance acts as antiseptics: 0.2% phenol and other substances. That stops the spread of pathogenic agents.

Subsequently, as fish excrete ammonia (NH_3) directly into water; the ammonia is lethal to salmon even in tiny doses. This ammonia-reaching water undergoes a neutralization process called Ammonia-Oxidizing Bacteria (AOB):

$$NH_4^+ + 1.5O_2 \rightarrow NO_2^- + H_2O + 2H^+ + \text{Energy} \quad \textbf{(2)}$$

Furthermore, the toxic Nitrite should not be exceeded in water; thus, they confirm another method called Nitrite-Oxidizing Bacteria (NOB):

$$NO_2^- + 0.5O_2 \rightarrow NO_3^- + \text{Energy} \quad \textbf{(3)}$$

All these processes, and temperature optimization, are made to compensate for the salmon environment: 13-16°C; temperature, and 6 mg/L; DO **(*Dissolved Oxygen - Environmental Measurement Systems*, 2025).**

<u>Points of Weakness:</u>

<u>High Operational Cost:</u>

Throughout the implementation of the recirculation aquaculture technique, Atlantic Sapphire's Bluehouse makes water circulation every half hour. They utilize high-powered stainless-steel pumps of a model MPFR-I chopper pump **(*Atlantic Sapphire Silage Plant With BioChop System | Landia*, 2026)**.

These pumps are able to drive approximately 2,700 liters per second through galvanized pipes in the range between 150 mm and 900 mm. The aquaculture utilizes about 36 pumps; yet, the cost of each one is 1,279.37 dollars **(*Centrifugal Pump Price List*, 2022)**. Each pump consumes about 150 kWh.

<u>Waste Disposal:</u>

The Atlantic Sapphire's Bluehouse utilizes the deep well injection system; that is, this is used to get rid of waste after the treatment process. The facility relies entirely on the capacity of the boulder zone aquifer—located 3,000 feet beneath the Earth's surface.

Since the water is recycled every half hour at the facility, the boulder zone accepts millions of gallons of treated effluent daily **(Bluehouse Salmon, 2025).** Due to this high amount of disposal, the infrastructure of Atlantic Sapphire's Bluehouse requires continuous maintenance. This helps to avoid the decreasing of the zone porosity.

Points of Strength:

Water Dependence:

The Atlantic Sapphire's Bluehouse demonstrates a highly effective self-contained water treatment system. Evidently, its water courses are divided into two categories: 5% freshwater and 95% saline water **(Bluehouse Salmon, 2025).** This water goes through a complex treatment process every half hour—optimizing the nitrate, DO, and other parameter levels. Moreover, it elevates the temperature. Then, after 10 days, they refill the 36 massive concrete tanks with 454,000 gallons each. The tanks are filled via high-powered stainless-steel pumps of a model MPFR-I chopper pump. Aquaculture pump consumes about 8.8-10 kWh/kg smolt with an average cost of $0.12 per kWh **(*This Is How Much Electricity Atlantic Sapphire Used per Kilo of Salmon*, 2021).**

Biosecurity and Disease Mitigation:

The Atlantic Sapphire utilizes thirty-six massive tanks to raise salmon fish. The aquaculture system utilizes an extensive approach that includes disinfection and nitrification. Even more evidently, the water intake is subjected to UV irradiation and ozone treatment before it ever contacts the fish **(Recirculating Aquaculture Salmon Network, 2025).**

This rigorous protocol eliminates the need for hazardous chemicals, allowing the facility to operate without antibiotic utilization. Ultimately, the viable bacteria drop significantly after this process to a range of 10^2 to 10^3 CFU/mL—creating a more sustainable clean environment for the salmon fish.

Electric Efficiency:

The Atlantic Sapphire's Bluehouse operates as a high-density land-based facility, maintaining a stocking density of 50 kg/m^3. Thus, it has the potential to produce nearly 9,500 tons of salmon, since the density of the salmon fish is 50 kg/m^3. This density is considered double the standard density of traditional open-ocean nets. That's why this exhibits approximately 0.34% of the global market **(Annual Report, 2022)**.

While aquaculture consumes 8.8-10 kWh/kg, the average cost is about 0.12 dollars per kWh **(*This Is How Much Electricity Atlantic Sapphire Used per Kilo of Salmon*, 2021).** Thus, the Bluehouse ensures a stable consumption of energy and production of fish.

Azahir Farm, El Beheira Government, Egypt

Overview

Figure 1.21: Azahir Farm

The agriculture sector faces the daunting challenge of providing adequate food and other necessities for a growing world population, which is projected to be nine billion by 2050. There is limited scope for the expansion of arable land, and the emerging threat to agriculture from climate change in the form of unpredictable weather, floods, and other disastrous events makes the task of providing enough food for the global population even more challenging **(Clements et al., 2011).** Since 1950, agricultural researchers have applied the scientific principles of genetics and breeding in an aggressive effort to improve crops grown primarily in less developed countries. Agriculture is understood to be the cultivation of plants that serve as food, provender, oils, and industrial products. This activity involves the use of different groups of plants: grasses, vegetables, flowers, legumes, oil-seeds, industrial crops, medicinal plants, and others. One of the most significant difficulties faced by designers of greenhouses is the construction of a structure that allows the correct regulation of the environmental conditions. Azahir Farm is the best farm for greenhouse in Egypt **as shown in the Figure(1.21).**

Mechanism

The mechanism of a traditional manual greenhouse relies fundamentally on passive thermodynamics and manual human intervention rather than automated control systems. Structurally, it consists of a skeletal frame—often made of wood, galvanized steel, or PVC pipes—covered with a transparent or translucent material such as polyethylene film or glass. The primary scientific principle at work is the "greenhouse effect." Short-wave solar radiation passes through the transparent covering and is absorbed by the plants and soil inside. These surfaces then re-radiate this energy as long-wave infrared thermal radiation, which cannot easily escape back through the covering, effectively trapping heat and raising the internal ambient temperature.

Environmental regulation in this system is entirely manual. For temperature and humidity control, farmers must physically open side vents, roof panels, or doors to allow cross-ventilation when the internal climate becomes dangerously hot or humid. Shading is achieved by manually deploying physical shade nets over the roof to block a percentage of incident sunlight during peak summer months. Irrigation is conducted using open-loop methods such as manual hose watering or manually operated drip lines, without the use of soil moisture sensors. Plant health and conditions are monitored exclusively through human visual inspection, meaning there are zero electronic sensors, microcontrollers, or feedback loops involved in the process.

Points of Weakness

Limited Repair:

One of the weaknesses of the traditional manual greenhouse is its limited repair flexibility. These devices are designed as compact and integrated systems that do not allow easy repair because it is difficult to access the internal components. When a sensor or circuit fails, the entire component needs to be replaced by technicians. Also, this increases the maintenance cost and affects its performance. As a result, while the device has high precision, its complex maintenance requirements make it less suitable for some laboratories.

Power Supply & Environmental Conditions:

The performance of the multi-parameter water depends on the stable power supply. Since the sensors and circuits that need electric input, any fluctuation in voltage can affect the accuracy of the readings. Also, variations in temperature and humidity can interfere with the electrochemical sensors, causing signal noise. This makes this device not suitable for many laboratories. The high dependence on a constant power source which limits the usage in testing areas. While the device has high precision, its sensitivity to environmental and electrical changes weakens.

Lack of Recycled Issues:

Scientific environments like not-recycled greenhouse require a lot of technological machines to provide farmers with the opportunity to conduct scientific experiments and analyse the results for a better quality of production. These smart greenhouse projects are significantly above the country and farmer's budget. For instance, the greenhouse requires some equipment, such as sensors and expensive structure, which provides the chance for innovative farmers and governments to apply their creative designs, projects, and prototypes to the real world. This is an example of only structure of the smart greenhouse not recycled. As same as smart greenhouse, other smart agriculture, also need some technological devices, which is used in maintaining production. Similarly, these are high-cost devices, highlighting the same issue.

Points of Strength

Cost Efficiency:

The greenhouse project provides a way to design and build a highly effective solution at a fraction of the cost of commercial systems. This project attracts more international and local investment, which is estimated in the millions. Greenhouse projects save money for Egypt because they are made from recycled materials, and agricultural instruments are often expensive and difficult to maintain. Also, it is useful for citizens to start a new job. The greenhouse project considers generating jobs like separating electronic components from recycled materials, project development, manufacturing, and sales. It offers opportunities to workers possessing various skills, including engineers, electricians, and other workers. Additionally, when the project becomes available at a low cost and small scale, farmer students can learn how it works, and technicians can build.

Enhancing Agriculture:

The implementation of an automated closed-loop greenhouse control system allows for monitoring of parameters, such as soil moisture, humidity, temperature, and light. By utilizing real-time measurement feedback, the system triggers immediate corrective actions through specific actuators. This adaptive mechanism effectively prevents sudden environmental fluctuations, thereby restoring and maintaining the optimal conditions required for sensitive and productive crops. By build this project with optimal environmental conditions to support agricultural production. This system will make agricultural food available in all seasons, which will increase the economy of Egypt.

Reducing Waste:

This project helps reduce pollution and waste by reusing old plastic and components to build new projects. Every year, millions of electric and electronic items are discarded. One of the main benefits of recycling is that it can reduce pollution because it does not release any emissions into the air. Using recycled materials in this project will save money for the government, instead of being thrown away or burned, and will release toxic substances. Recycling old things protects public health by reusing and repurposing components like motors, sensors, and circuit boards. Repurposing old plastics for the structure and salvaging electronic components will minimize e-waste generation and prevent these hazardous materials from entering the environment.

Jain Irrigation Systems

Overview

The **Jain Irrigation Systems** drip irrigation technology was originally developed in **India** to help farmers manage water more efficiently, especially in regions where water resources are limited. The company, founded in **1963**, focused on creating **micro-irrigation systems** that deliver water directly to plant roots through small tubes and emitters. One of the well-known systems developed by the company is the **Jain Turbo Excel Dripline**, which distributes water slowly and evenly across crops.

The mechanism of early irrigation depended on flood or surface irrigation, resulting in an enormous waste of water; moreover, there was a probable wasting of plant species due to the overload of water on the roots. Consequently, **Jain Irrigation Systems provided a drip irrigation (Jain Turbo Excel Dripline) by delivering drops of water directly into the root of the plant, through specific pressure-regulated emitters**. So, this method significantly reduces water loss caused by evaporation and runoff. In addition,

Jain irrigation systems allow the integration of **fertigation**, in which the fertilizers are mixed with the water in a specific tank, and then the water is transferred through the pipes to the roots in the form of drops, allowing the falling of an appropriate amount of fertilizers, which is a sufficient amount for the roots. In addition, the Jain irrigation system has a replacement for the old pipelines, which were basic and needed manual adjustments. The new pipe had been durable. Also, the control systems had been improved by **motor-controlled or pressure-compensated drippers**, allowing extremely accurate and repeatable water distribution, eliminating manual intervention, and enhancing irrigation speed and consistency.

Mechanism

Firstly, water is stored in a tank of capacities ranging from 500 liters to several thousand liters, depending on the size and the area of the farm. So, as the fertilizers are mixed with water. So, the concentration of it in water should be embedded in the range of concentration 1-3 grams per liter.

And then, water flows through a very thin pipe of a diameter ranging between 25-50 mm, which branches into lateral pipes (16-22 mm diameter), that carry water to the drippers distributed across the field or greenhouse.

The water enters pressure-compensated drippers, which release water at rates of 2-4 liters per hour. These drippers automatically adjust to pressures between 0.8 and 3.0 bar, ensuring a uniform delivery across the long pipes of length up to 500 meters. After the water exits from the drippers, it infiltrates a soil of depth 10-15 cm, providing an optimal moisture zone for most vegetable and greenhouse crops.

So, on the other hand, the integrated fertigation system must be a shared factor and step, because it works on the dissolving of nutrients in water, delivering them precisely to plant roots. Its efficiency is about 90-95%. In advanced systems, soil moisture sensors work on monitoring the water content and automatically trigger irrigation when the moisture drops below 15–20%, maintaining ideal soil conditions without manual intervention.

Durable pipelines and pressure-regulated drippers are designed to resist clogging and maintain a flow rate within ±5% of the designed output over long-term use. The system also allows for flushing and maintenance, which prevents sediment from building up and ensures consistent performance. By delivering precise amounts of water and nutrients, the system reduces water waste by 40–70% compared to traditional flood irrigation, moreover, improves crop yield by 20–30%, and minimizes labor requirements.

Points of Weaknesses

Limited Repair:

One of the weaknesses of the traditional manual greenhouse is its limited repair flexibility. These devices are designed as compact and integrated systems that do not allow repairing anything easy because it difficult to access the internal component. When sensor or circuit fails, the entire component needs to be replaced by technicians. Also, this increases the maintenance cost and effect on its working. As a result, while the device has high precision, its complex maintenance requirements make it less suitable for some laboratories.

Lack of Recycled Issues:

Scientific environments like non-recycled greenhouses require a lot of technological machines to provide farmers with the opportunity to conduct scientific experiments and analyse the results for a better quality of production. These smart greenhouse projects are significantly above the country and farmers' budget. For instance, the greenhouse requires some equipment, such as sensors and an expensive structure, which provides the chance for innovative farmers and governments to apply their creative designs, projects, and prototypes to the real world. This is an example of only the structure of the smart greenhouse not recycled. As same as smart greenhouse, other smart agriculture systems also need some technological devices, which is used in maintaining production. Similarly, these are high-cost devices, highlighting the same issue.

Dependence on Agriculture and Weather Conditions

Another weakness of Jain Irrigation Systems Ltd. is that much of its business depends on the agricultural sector. Agriculture is strongly affected by weather conditions such as droughts, floods, and irregular rainfall. When farmers face difficult weather conditions or poor harvests, they may delay or reduce their investment in irrigation systems and farming technology. This can directly affect the company's sales and revenue. Because of this dependence on agriculture and climate conditions, the company's performance may fluctuate from year to year. This makes the business less predictable compared with companies that operate in industries that are less affected by environmental factors.

Points of Strength

Enhancing Agriculture:

The implementation of an automated closed-loop greenhouse control system allows for monitoring of parameters, such as soil moisture, humidity, temperature, and light. By utilizing real-time measurement feedback, the system triggers immediate corrective actions through specific actuators. This adaptive mechanism effectively prevents sudden environmental fluctuations, thereby restoring and maintaining the optimal conditions required for sensitive and productive crops. By building this project with optimal environmental conditions to support agricultural production. This system will make agricultural food available in all seasons, which will increase the economy of Egypt.

Leadership in Micro-Irrigation Technology

Jain Irrigation Systems Ltd. has built a strong reputation as one of the world's leading providers of micro-irrigation solutions, especially drip and sprinkler irrigation systems. The company focuses on technologies that help farmers use water more efficiently while improving crop productivity. By developing advanced irrigation equipment and providing technical support to farmers, Jain Irrigation has become a trusted partner in modern agriculture. Its systems help reduce water waste, increase crop yields, and support sustainable farming practices. This technological leadership gives the company a strong competitive advantage in global agricultural markets where water conservation and efficient food production are becoming increasingly important.

Commitment to Research, Innovation, and Sustainability

A key strength of Jain Irrigation Systems Ltd. is its strong commitment to research, innovation, and sustainable agriculture. The company invests significantly in research and development to improve irrigation technologies and agricultural practices. Its innovations focus on water conservation, energy efficiency, and increased crop productivity. Jain Irrigation also promotes sustainable farming by educating farmers on efficient water management and modern cultivation techniques. This dedication to innovation allows the company to stay competitive in a rapidly evolving agricultural industry. By combining technology with sustainability goals, Jain Irrigation supports global efforts to address water scarcity and ensure long-term food security.

Automated Aquarium Control Systems

Overview

An experimental study focused on developing and evaluating an automated, microcontroller-based system designed to monitor and regulate the pH, temperature, and turbidity of aquarium water, specifically catering to the needs of ornamental fish cultivation. Recognizing that maintaining optimal water quality is often a time-consuming challenge for fish breeders ,which can inadvertently lead to poor water conditions, stunted growth, metabolic issues, and even fish mortality. The researchers engineered a comprehensive solution using an Arduino Uno as the central controller. The automated system seamlessly integrates an E-201-C pH sensor paired with solenoid valves to automatically dispense pH-up or pH-down solutions, ensuring the water's acidity remains stable within the ideal range of 7 to 7.5. Additionally, a DS18B20 temperature sensor is utilized to constantly measure thermal conditions, automatically triggering either a heater to warm the water or a cooling fan to lower the heat, thereby maintaining a consistent and healthy water temperature between 24 and 27 degrees Celsius. **This shown in Figure (1.22)** To address water clarity, a turbidity sensor continuously checks for suspended particles and dirt; if the water's turbidity exceeds the acceptable threshold of 10 NTU, the system activates water pumps to automatically drain the dirty water and replace it with fresh,

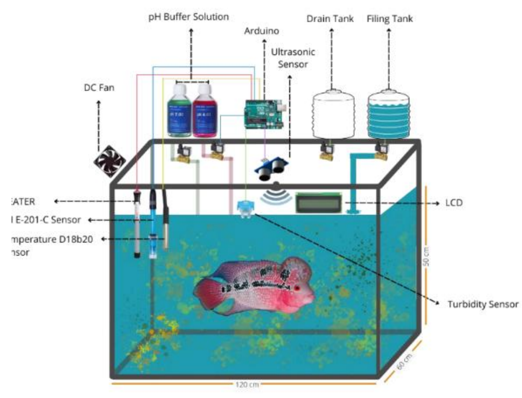


Figure 5.22: The parts of the system

clean water, utilizing an ultrasonic sensor to precisely monitor and control the water level during this entire exchange process. To thoroughly validate the effectiveness of this automated setup, the researchers conducted a comparative 10-day experiment, observing it alongside a conventional aquarium lacking any control mechanisms. The results overwhelmingly demonstrated the automated system's superiority, as it successfully maintained all targeted water parameters within their optimal ranges without requiring manual intervention, whereas the uncontrolled aquarium experienced significant fluctuations, resulting in acidic, cloudy, and potentially harmful water conditions. Ultimately, this study proves that implementing such an automated monitoring and control system provides a highly efficient, reliable, and low-maintenance method for preserving pristine water quality, thereby significantly promoting the health, growth, and overall well-being of ornamental fish.

Mechanism

The innovative automated control system for the ornamental fish aquarium relies on a highly precise and integrated mechanism that operates continuously around the clock to ensure an optimal and healthy aquatic environment for the fish. The Arduino Uno microcontroller acts as the electronic mastermind that receives real-time data from three advanced main sensors, analyzing it in fractions of a second to make the necessary automated decisions to operate auxiliary equipment without any human intervention. Meanwhile, it clearly displays all vital readings on a dedicated screen to enable the user to continuously monitor the overall condition of the aquarium. This dynamic mechanism begins with strict temperature control via a highly accurate thermal sensor fully immersed in the water to continuously measure thermal changes. The central system is programmed to maintain an ideal temperature range precisely between 24 and 27 degrees Celsius to suit sensitive ornamental fish. If the sensor detects a drop in temperature below 24 degrees Celsius, the Arduino sends an immediate electrical signal to turn on a water heater to gradually warm the water and raise the temperature.

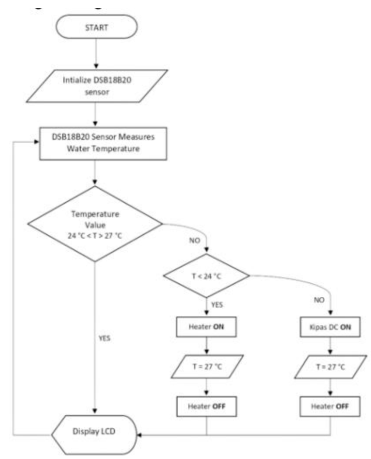


Figure 1.63: Temperature sensor software design.

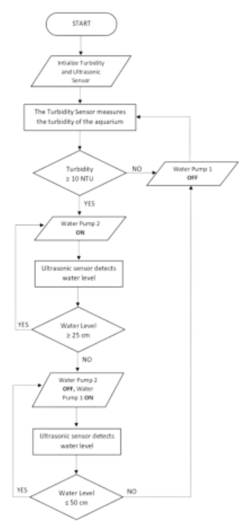


Figure 1.24: Turbidity sensor software design

Conversely, if the temperature rises and exceeds the 27-degree Celsius barrier due to external weather factors, the

Figure 1.25: pH sensor software design.

system quickly intervenes by operating a dedicated cooling fan to dissipate excess heat and cool the water surface. Once the temperature returns to stability within the target range, these devices stop automatically. Secondly, the system continuously monitors water clarity to eliminate the accumulation of toxic waste and food debris using a special turbidity sensor . The system considers the water pure and healthy as long as the turbidity level is below 10 standard units (NTU). However, once this percentage exceeds the permissible limit and the water becomes cloudy, the system immediately recognizes a pollution hazard and activates a powerful drain pump to suction and remove the turbid water from the aquarium. Simultaneously with this extraction process, an advanced ultrasonic sensor mounted at the top of the aquarium accurately monitors the rate of water level drop.

When the water level drops and reaches exactly 25 centimeters, the drain pump stops immediately to protect the fish from aquarium dryness and leave a safe space for them, simultaneously triggering another fill pump to pump clean and pure water from a connected reserve tank. The active filling process continues until the water level rises back to the 50-centimeter mark, at which point the pump stops automatically and the water change cycle is completed. Thirdly, the system handles a crucial chemical task represented in adjusting the pH level using a precise sensor that continuously reads the acidity and alkalinity of the water to maintain it within the moderate and ideal range between 7 and 7.5. If the reading drops below 7, indicating an increase in acidity levels that could be fatal to the fish, the system opens an electrical solenoid valve to automatically pump a calculated amount of alkaline solution to raise the level and neutralize the water. If the reading rises above 7.5, another electrical valve is opened to pump an acidic solution to lower the level. To ensure the accuracy of this sensitive interactive process and avoid adding excessive amounts of chemical solutions that might shock the fish, the system intelligently pauses the pumping process for a full five minutes after each addition. This pause allows the added solution to spread and mix completely and homogeneously with every drop of the aquarium water. After these five minutes have passed, the sensor resumes taking a new reading to assess the situation and ensure the pH has stabilized within the safe range; if necessary, the process is carefully repeated. All these complex physical and chemical processes integrate into an endless and continuous cycle, creating a stable and reliable ecosystem that protects the ornamental fish and ensures their flourishing.

Points of Weakness

High Initial Setup Costs and Technical Complexity

Despite its benefits, a notable negative consequence of this automated system is the high initial setup cost and the advanced technical complexity required for its assembly. Unlike a traditional aquarium setup which only requires a basic tank, a simple filter, and a standard heater, this advanced system relies on an array of electronic components. The user must purchase an Arduino Uno, specialized sensors (pH E-201-C, DS18B20 temperature, turbidity, and ultrasonic), multiple water pumps, solenoid valves, cooling fans, and LCD screens. Beyond the financial investment, assembling these parts requires a solid understanding of electronics, wiring, and computer programming to calibrate the microcontroller correctly. For the average fish hobbyist or small-scale breeder, this steep learning curve and the high upfront financial barrier can be highly discouraging, making the technology inaccessible to those without a technical background or a large budget.

Vulnerability to Power Outages and Hardware Failures

A critical vulnerability of this highly automated setup is its absolute dependence on a continuous electricity supply and the severe risks associated with hardware malfunctions. Because the entire ecosystem is controlled electronically, a sudden power outage would immediately disable the heaters, cooling fans, monitoring sensors, and water pumps, rapidly leading to a deteriorating environment that could harm the fish. Furthermore, electronic sensors are prone to occasional glitches or failures. If the pH sensor malfunctions and provides a falsely high reading, the Arduino might continuously pump acidic pH-down solution into the tank without stopping. Because the system operates independently without human supervision, such a catastrophic technical failure could easily poison the entire tank before the owner even realizes there is a problem, resulting in the rapid and total loss of the aquatic livestock.

Ongoing Machine Maintenance and Sensor Calibration

Finally, while the system drastically reduces the manual labor of changing water and testing parameters, it introduces a completely new and potentially frustrating requirement: ongoing machine maintenance and frequent sensor calibration. Electronic water sensors, particularly pH and turbidity probes, do not remain perfectly accurate indefinitely. They are prone to biological fouling, algae buildup, and calibration drift. Therefore, the owner must regularly remove, clean, and manually recalibrate these delicate sensors using specialized buffer solutions to ensure the system continues to make accurate decisions. Additionally, the system is not entirely "hands-off"; the user must diligently monitor and manually refill the external reservoirs holding the clean fresh water and the chemical pH modifiers. If these reservoirs run dry, the automated system becomes useless. This shifts the burden from traditional aquarium maintenance to complex equipment maintenance, which can be equally demanding.

Points of Strength

Enhanced Fish Health and Reduced Mortality Rates

One of the most significant positive consequences of this research is the dramatic enhancement of fish health and the substantial reduction in mortality rates. Ornamental fish, such as the Louhan, are highly sensitive to environmental fluctuations. By utilizing an automated system that strictly maintains the pH between 7 and 7.5, the temperature between 24 and 27 degrees Celsius, and turbidity below 10 NTU, the aquarium simulates a perfectly stable natural habitat. This constant environmental stability eliminates the physical stress that fish typically endure during sudden changes in water chemistry or temperature. Consequently, the fish are far less susceptible to metabolic disorders, stunted growth, and common aquatic diseases. For breeders, this translates to a higher survival rate, better coloration, and more successful breeding cycles, ultimately raising the overall quality and commercial value of the ornamental fish being cultivated.

Significant Time and Labor Savings for Breeders

Another major advantage is the immense amount of time and physical labor saved for both casual hobbyists and commercial fish breeders. Traditionally, maintaining an aquarium requires tedious daily or weekly routines, including manually testing the water with chemical kits, siphoning out dirty water, and carefully measuring and adding balancing chemicals. This automated microcontroller-based system completely takes over these demanding chores. With the Arduino Uno constantly processing sensor data and independently activating water pumps, heaters, cooling fans, and solenoid valves, the human element is largely removed from the daily maintenance equation. Busy individuals who previously lacked the time to properly care for demanding ornamental fish can now easily maintain pristine tanks. This level of automation transforms fish keeping from a labor-intensive chore into a stress-free, highly manageable hobby or business operation.

Improved Water Conservation and Resource Efficiency

The implementation of this automated system also promotes improved resource management, specifically regarding water conservation and the efficient use of chemical treatments. In traditional aquarium maintenance, owners often perform large, arbitrary water changes on a set schedule, regardless of whether the water is actually excessively dirty, leading to unnecessary water waste. This automated system, however, relies on a highly accurate turbidity sensor that only triggers a water replacement cycle when the dirt levels explicitly exceed the 10 NTU threshold. Furthermore, the ultrasonic sensor ensures precise control over the exact volume of water drained and refilled. Similarly, the targeted release of pH-up and pH-down solutions via solenoid valves—combined with the programmed five-minute resting periods—prevents the excessive application of chemicals. This precision minimizes waste, making the system both economically efficient and environmentally friendly over the long term.

Chapter 2

Generating and defending a solution

Solution requirements

Cost Efficiency:

The project provides a way to design and build a smart recycled greenhouse at low cost. This project attracts more international and local investment, which is estimated in the millions. Smart greenhouses save money for farmers and governments because they are made from recycled materials, and non-recycled smart greenhouse instruments are often expensive and difficult to maintain. Also, it is useful for citizens to start a new job. Smart greenhouse project considers generating jobs like separating electronic components from recycled materials, project development, manufacturing, and sales. It offers opportunities to workers possessing various skills, including engineers, electricians, and other workers. Additionally, when the project becomes available at a low cost and small scale, students and farmers can learn how it works and technicians can build.

High Precision and Accuracy:

One of the advantages of the closed-loop smart greenhouse is its high precision and accuracy in monitoring and controlling environmental parameters. Plant growth is highly sensitive to temperature, humidity, soil moisture, and light intensity, which affect plant health and productivity if small measurement errors occur. Therefore, calibrated sensors are essential to provide reliable real-time data. High precision also improves data analysis quality, allowing meaningful correlations between environmental variables and plant growth. Consequently, precision and accuracy directly enhance system performance, reliability, and scientific validity.

Safety:

In the process of constructing the prototype, team members should wear lab coats and lab safety glasses to protect themselves from any dangerous power that are used in implementing the prototype. In addition, they must wear protective gloves to keep their hands safe when they use any dangerous tools or when dealing with electricity. They also should put face covers to save their faces from the heat or fire when they are putting the metal parts of the prototype together.

Eco-friendliness:

This project helps reduce pollution and waste by reusing old tools to build new ones with the same or different functions. Every year, millions of electric and electronic items are discarded. One of the main benefits of recycling is that it can reduce pollution because it does not release any emissions into the air. Using recycled materials in this project will save money for the school, instead of being thrown away or burned release toxic substances. Recycling old things protects public health by reusing and repurposing components like motors, sensors, and circuit boards. This project helps minimize e-waste generation and prevents these hazardous materials from entering the environment.

Design Requirements

To determine whether the solution works successfully or not, it should compensate for several requirements:

The prototype scale:

The prototype should be constructed with an area that doesn't exceed $0.25m^2$ for the crop growing area. The housing is preferred to be built utilizing natural materials.

Parameters:

The prototype should be able to monitor at least three parameters. These parameters include one of the following: temperature, pH, or salinity. Moreover, the sensors used should be calibrated according to a scientific paper that specifies the conditions needed for whatever grows inside the prototype.

Feedback System:

The prototype should work continuously for approximately 6 hours. Throughout, it would provide the measure of the three selected parameters, offsetting the decrease/increase. The entire data would be collected by sensors to Arduino, ensuring a quick synchronization.

Controller Algorithm:

The prototype should have at least one control strategy, such as on/off control, rule-based expert logic (IF–THEN rules), or adaptive or scheduled control algorithms.

Selection of Solution

The solution chosen, Smart Recycled Greenhouse, is focused on providing a sustainable, controlled agricultural system. The prototype will be approximately 0.25 square meters in size, made from recycled materials like recycled wood and transparent plastic bottles to maintain low environmental impact and cost while allowing for passing sunlight into the greenhouse.

The control method is a closed-loop adaptive feedback system, which allows monitoring of environmental conditions and adjustment for any deviations outside established optimal ranges. The system will monitor soil moisture, air humidity, and light intensity. These three parameters will be measured with the assistance of capacitive soil moisture sensors, the DHT22 sensor, the turbidity sensor, and either the LDR light sensor or the BH1750 light sensor.

Data from these different sensors will be processed by an Arduino. A microcontroller that has been programmed with Rule-Based IF-THEN Logic prevents the rapid switching on/off of actuators. For example, once soil moisture falls below a threshold level, the moisture will trigger a Drip-Irrigation Pump to deliver additional moisture. Humidity levels will trigger the Ultrasonic Mist Maker to add humidity, and Ventilation Fans will trigger to increase airflow. Additionally, servo-controlled shading will be applied when sunlight exceeds optimum light intensity levels.

After choosing the solution, the practical implementation of the solution includes all of the sensors and actuators integrated and protected from moisture damage and electrical hazards in a closed system. The system will run automatically for at least six hours and collect information on the environment at regular intervals to measure performance, energy efficiency, and stability of plant growth.

Sustainable materials are an important part of the structural design of this solution. Recycled wood can be used for the frame, and clear plastic bottles will allow enough light to pass through the system. The power supply for the system is a regulated DC supply that also has a protective circuit designed to prevent pump dry-running and accidental electrical fault. The overall integrated design demonstrates the effectiveness of this smart agriculture solution as a low-cost.

Selection of Prototype

The project is a Smart Recycled Greenhouse, as illustrated in **Figure (2.1)**. The Smart Recycled Greenhouse is able to optimize the environmental conditions for crops, including strawberries and orchids. This is via utilizing an adaptive feedback system gathering its results from several calibrated sensors. In order to monitor three parameters: soil moisture, air humidity, and light intensity. The entire system works as follows:

Figure 2:1: The 3D modeling

First of all, the prototype measures the tracked water level within the greenhouse air via a DHT 22 sensor. As this sensor mainly depends on moisture-absorbent dielectric materials, when the moisture increases, the dielectric material absorbs this vapor; thus, the dielectric constant varies and the capacitance in turn. Then, it measures the moisture in the soil itself via a capacitive soil moisture sensor. The sensor operates using a similar mechanism.

Continuous monitoring ensures the optimal conditions for the photosynthesis process, which is considered a main source of food for plants. Typically, the BH1750 sensor is composed of a semiconductor material with an inverse-proportionality to light intensity. As the resistance decreases when the photons hit the sensor; this variation in resistance is a fluctuating analog voltage to determine if it is sunny, cloudy, or dark.

After that, the sensor should be calibrated to the standard environment required by each crop individually: strawberries and orchids. Furthermore, the whole system is automated using an Arduino; that is, it can maintain the optimal environment required by strawberries and orchid respectively. The prototype would be constructed with an area that doesn't exceed $0.25m^2$—0.5 m for both the length and width.

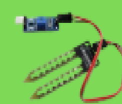
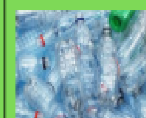
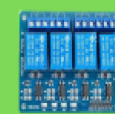
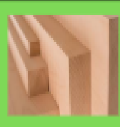
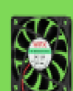
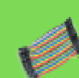
Chapter 3

Constructing and Testing a Prototype

Materials and Methods

Materials:

Table 3.2: Materials of construction

Materials	ESP 32	LDR Sensor	DHT-22 Sensor	Soil moisture sensor	Recycled- Bottles	Relays
Description	The controller board that used for controlling Feedback system	used to detect the light and gives feedback for ESP 32	used to measure temprature and Humidity	Used to measure the humidity of Soil	Used a transparent cover to allow sunlight passing	Used to control the operation of Actuator
Quantity	1	1	1	1	30	1
Image						
Materials	Beech Wood	Fans	Nails	White Pumps 5V	Power Supply	Jumper wires
Description	to construct the frame of our Prototype	to cool weather condition around plant	to connect the frame of prototype	to pump water inside prototype	to emerge power inside the circuit	used to conncet components of circuit
Quantity	8 meters	3	50	2	1	60
Image						

Methods:

1. **Prototype design and sketching:** The project started with visualization and selecting the dimensions to make a 3D model using SolidWorks, as shown in **Figure 3.1.**

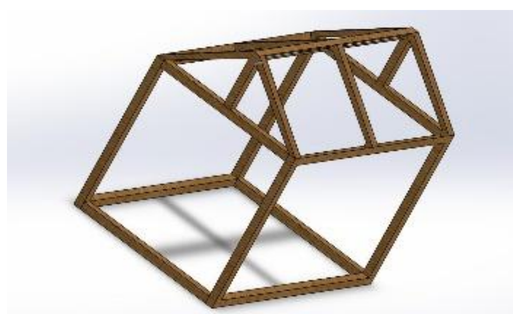


Figure 3.7: 3D Design

2. **Electronic System Integration:**
The control system was built using an ESP32 microcontroller mounted on a breadboard, including relay modules to safely handle high-voltage components, alongside a 12V power supply to operate actuators. All components were interconnected with ESP32, as shown in **Figure 3.2**, ensuring a robust electronic architecture for the automated operations.

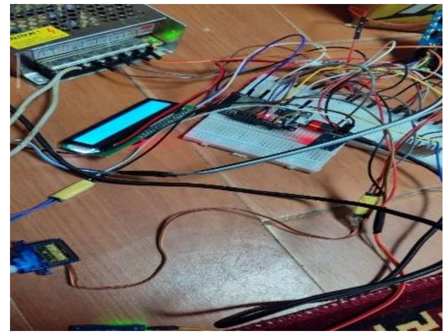


Figure 3.8: Electronic System

3. **Sensors and actuators integration:**
A DHT22 sensor was connected to monitor temperature; the ESP was programmed to activate a cooling fan when the temperature threshold was exceeded. Regarding the soil moisture sensor, it was utilized to activate a secondary water pump until the soil took the required water, as shown in **Figure 3.3**. Also, a Light Dependent Resistor (LDR) was used to monitor light intensity to control a 220V to provide artificial lighting, while a DC motor driven by an H-bridge module was implemented to operate a shading system.

Figure 3.9: Sensors & Actuators

4. **Programming system:**
An ESP32 code was programmed using Arduino IDE to control sensors and actuators, as shown in **Figure 3.4**. The system continuously reads data and activates corresponding actuators to maintain optimal environmental conditions without human intervention.

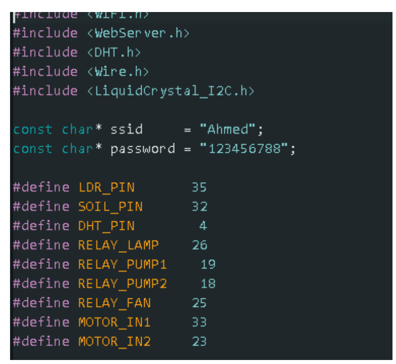


Figure 3.10: Programming

5. **System testing and calibration:**
To ensure the accuracy of the data collected from sensors, the calibration was conducted. The temperature data from the DHT22 sensor was validated by comparing the reading against an external digital thermometer. For the soil moisture sensor, readings were recorded in both completely dry and fully saturated soil to handle the water pump. This physical calibration ensured that the logic responded accurately to actual environmental conditions.

6. **Web-Based Monitoring and Control:**
   A web-based interface was integrated and connected to the ESP32 via Wi-Fi. This website allows real-time monitoring of parameters and data, and allows manual control of the system components, which provides flexibility with automated operation and emergency cases, as shown in **Figure 3.5**.

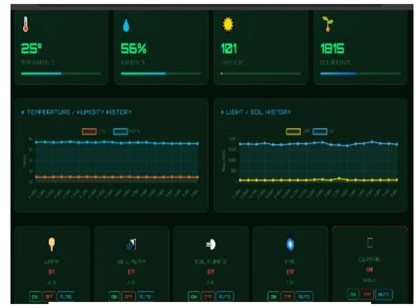


Figure 3.11: Website control

7. **Assembly of the whole prototype:**

   A wooden frame was constructed using a combination of fabricated and recycled materials to form a box structure as a container for soil, as shown in **Figure 3.6**. Also, recycled plastic bottles were flattened to serve as a cover.


Figure 3.12: Final Assembly

<u>Safety precautions</u>

Since safety is the most important requirement in constructing the prototype, safety precautions were taken as a primary concern. Whereby, gloves and lab coats were used to avoid any damage and negative consequences. Because most of the construction involved creating electric circuits using a soldering iron to connect different components, insulator gloves were used to avoid any electric shocks and any damage caused by the heat of the soldering iron. Furthermore, the prototype was constructed away from any flammable materials to avoid any fires. Safety precautions were. Moreover, certified eye goggles were used to protect our eyes from high-intensity laser beams. When the wooden structure that holds the circuits and the other components was being built, caution was taken while using any sharp tools. Some emergency tools, such as a fire extinguisher, were brought in during construction to nip any abrupt dangers in the bud.

<u>Test Plan</u>

A comprehensive set of criteria for testing was established to ensure the feasibility of the prototype to meet design requirements:

1. The prototype was tested to ensure continuous operation for at least 6 hours without human intervention. The structure was verified to be assembled from recycled components.
2. The parameters were monitored to ensure accurate sensor readings within a defined range. Sensors were calibrated using standard methods to ensure that accuracy does not exceed the ±5%.
3. The ESP32 controller was tested to ensure the correct implementation of programming logic like IF-THEN rules, and the system was tested under different conditions to evaluate its ability to detect changes.
4. The system was tested to ensure stable power, voltage, and reliable performance. Also, the complete documentation, including circuit diagrams, was tested.

Data Collection

Regarding the collection of data for the capstone project, a systematic set of measurement tools with high accuracy was utilized to ensure the reliability of the smart greenhouse's environmental control system. Specifically, for the thermal and atmospheric regulation, the ambient temperature and humidity were continuously recorded using the DHT22 sensor, which inherently provides readings with an accuracy of ±0.5°C and ±2% RH. To verify the accuracy of these readings during testing, an external digital thermometer and hygrometer with a precision of [±0.1°C and ±1% RH] were used. Moving forward, the hydrologic assessment of the soil was conducted using the soil moisture sensor, which was calibrated and verified against a commercial soil moisture meter with a precision of [±1%]. For the photometric control system, light intensity was monitored using an LDR, and the actual illuminance was verified using a digital Lux meter with a precision of [±5 Lux]. Finally, to ensure the stability and safety of the 12V power supply and the system's actuators (cooling fan, water pump, and DC motor), the voltage and current were determined by a digital multimeter with a precision of [±0.01 V and ±0.01 A], respectively.

Results:

Negative Results:

Initially, the system had noticeable delay and unstable behavior in the circuit, such as vibration-like fluctuations. After testing, the issue was caused by the water pump because it was connected to 12 volts instead of 5 volts. The issue was solved by adding the voltage regulator to reduce the voltage to 5 volts.

The project was implemented using Arduino Uno as the controller. After connecting the circuit components, it became clear that the system required data monitoring on a website to make it easier for the user. So, the issue was solved by using an ESP32 instead of an Arduino Uno to allow integration with Wi-Fi.

Positive Results:

The sensor data indicates stable and controlled environmental conditions, with temperature, soil moisture, humidity, and light maintained within desired ranges. This confirms the effectiveness of the feedback control system as shown in blew Figures.

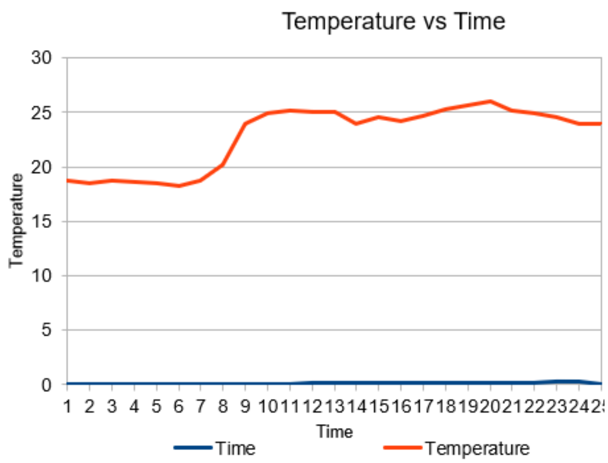


Figure 3.7: Temperature variation over time

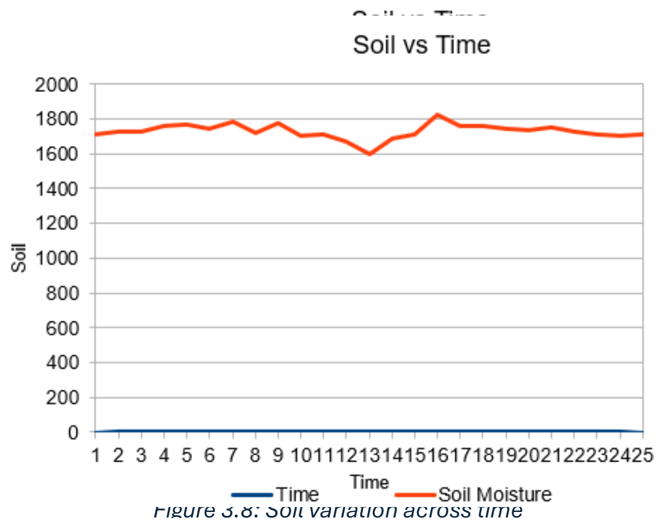


Figure 3.8: Soil variation across time

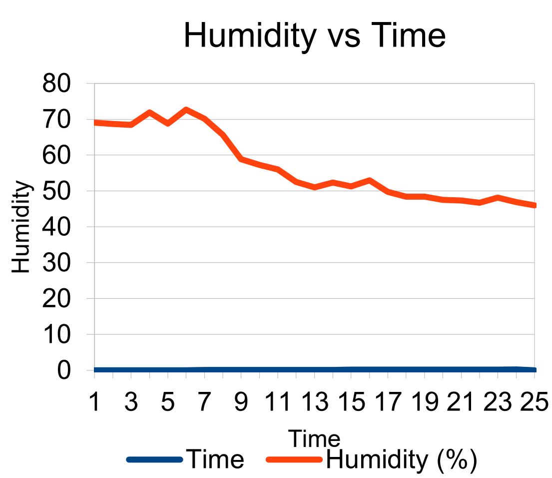


Figure 3.9: Humidity variation across time

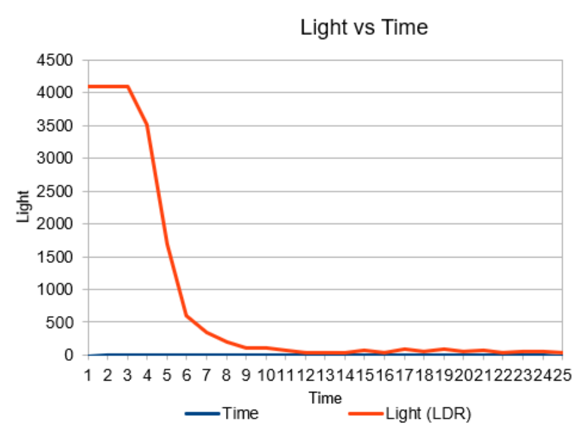


Figure 3.10: Light variation across time

Table 3.2: Parameters average value across each individual day for 5 consecutive days

Day Number	Temp	Humidity	soil	Light
Day 1	18.7	69.1	1713	4095
	25	52.5	1671	36
	22.2	65.7	1723	205
Day 2	18.2	68.5	1713	4095
	24.9	50.1	1671	36
	22	60	1723	205
Day 3	17.9	68.8	1718	4095
	24	52.2	1617	110
	20.5	50	1733	205
Day 4	18.9	69.1	1713	4095
	25.2	53.2	1671	32
	22.2	65.7	1723	205
Day 5	19	67	1728	4095
	25	52.5	1671	36
	23	65.7	1723	205

Chapter 4

Evaluation, Reflection, Recommendations, and Discussion

Analysis:

After constructing the small-scale prototype, a test plan was conducted to verify that the prototype satisfies the design requirements. These criteria included cultivation area, which was measured to ensure that it does not exceed 0.25 m^2, corresponding to dimensions of 0.4 m × 0.6 m. Moreover, the system's endurance was evaluated through continuous operation for 6 hours per day over 5 consecutive days to emphasize its reliability. This performance resulted from an integration of electric and mechanical concepts.

Structural Engineering

The primary purpose of this investigation is to validate the system's capacity to sustain an optimal agricultural microclimate amidst the harsh, desert-like conditions of the 6th of October City. To achieve this purpose, the project's performance is evaluated across several interdisciplinary aspects:

At the outset, the extreme environmental loads of the region—elevated temperature (approximately 36.0 °C)—dictate strict architectural constraints. That is why the structure must satisfy two critical criteria: passive air circulation and climate optimization. In response to these constraints, the **even-span greenhouse configuration** emerges as the optimum solution, **as illustrated in Figure (4.1).** It is a greenhouse structure composed of two equal roof slopes that accommodates two to three rows of plant benches.

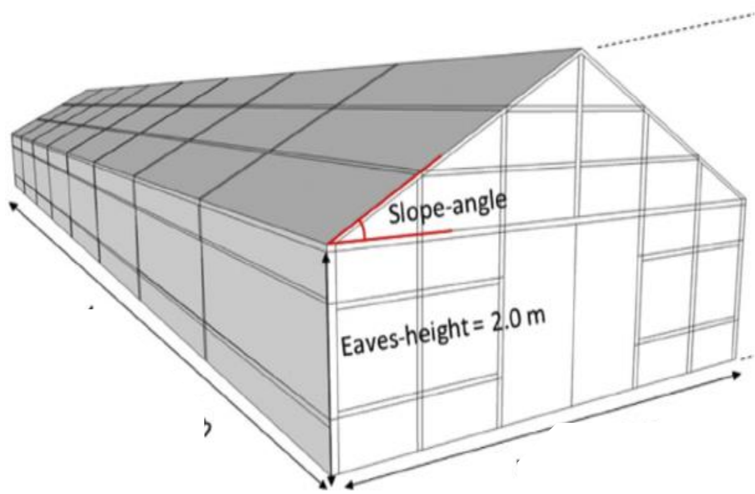


Figure 4.13: Even-Span Greenhouse Structure ***(Kwon et al., 2015)***

Under desert-like conditions, the even span relies on several approaches that enable it to gradually decrease the temperatures:

First of all, the number of spans enhances the microclimate by regulating the air temperatures via smooth air paths. The 30° angle of inclination based on Bernoulli's principle **(Equation 1)**:

Equation 1: Bernoulli's Principle

$$P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2$$

To validate the aerodynamic efficiency of the structure, based on the average air velocity of the city of 6th October, which is estimated at 6.0 m/s ***(Simulated Historical Climate & Weather Data for 6th of October City - Meteoblue, n.d.).*** Determining the structural geometry at a 30° roof and applying Bernoulli's principle resulted in a negative static pressure zone of 11.5 pascals. This pressure differential acts as a natural extraction system; air naturally flows from regions of higher pressure to regions of lower pressure. In light of these considerations, the quantified Air Changes per Hour (ACH), which was modeled utilizing **(Equation 2)**:

Equation 2: Air Changes per Hour (ACH)

$$ACH = \frac{Q_{hourly}}{V}$$

Where Q is defined as hourly airflow capacity per structural volume (V). This equation determines ventilation efficiency; the calculated airflow corresponds to 1.388 Air Changes per Hour.

Furthermore, the small-scale prototype has been constructed with a scale of 1:4.6 to the standard dimensions ***(Ponce et al., 2015)***; a width of 0.6 m, and a length of 0.4 m. Even more, the symmetric structure demonstrated **in Figure 4.1** assists in distributing the load across the structural frame, which resulted in enhancing the structure's resistance to wind (ideal yearlong utilization) ***(Admin, 2025).*** This behavior arises from the symmetrical roof, which redistributes the aerodynamic loads via the configuration of pressure along the entire foundation. Thereby, it prevents localizing the stress, enhancing the overall structural stability.

Soil-Water Dynamics:

This study addresses the limited soil quality for sustainable cultivation in a hyper-arid soil profile. The city of 6th October is characterized by extremely low soil moisture (approximately 1%); meanwhile, the coriander and orchid require high, consistent moisture for survival. To verify this criterion, some quantitative analysis is conducted.

Primarily, the particle density is considered, with an average value of 2.66 g/cm^3. This corresponds to a bulk density estimated at 1.4 – 1.8 g/cm^3 ***(University of Nebraska, 2026)***. These parameters will be utilized to calculate the total soil porosity **(Φ),** according to the following formula that was established by **Shreeja (2016) (Equation 3)**:

Equation 3: Soil Porosity

$$\Phi = \left(1 - \frac{\rho_b}{\rho_p}\right) \times 100$$

Where ρ_b is bulk density, and ρ_p is particle density. This equation determines the porosity of soil; it corresponds to a range between 32.3 and 52.6%, depending on soil compaction. This indicates a relatively high concentration of macro-pores, which can rapidly percolate water.

Within this project, the coriander typically occupies a shallow root zone of approximately 0.01-0.02 m, particularly in sandy soils, allowing roots to absorb water ***(Modhavadia et al., 2025)***. Thus, the amount of water that would be pumped via the prototype's environmental optimization system is calculated utilizing the following mathematical model **(Equation 4)**:

Equation 4: Volume of water required for soil

$$V_w = A \times Z \times (\theta_{FC} - \theta_i)$$

Where V_w is the total required water, Z indicates the effective root zone depth, and A is the prototype area. Furthermore, θ_{FC} and θ_i are the volumetric water content and the initial water content, respectively. The irrigation demand corresponds to approximately 0.5 L of water per irrigation cycle.

Feed-Back System:

The observed stability in the prototype results is explained by principles of thermodynamic equilibrium and hydrological balancing in the system. The smart greenhouse uses a feedback system dependent on the ESP32 microcontroller to monitor environmental variables within defined conditions against external action perturbations, which is described in a schematic diagram **as shown in Figure (4.2).**

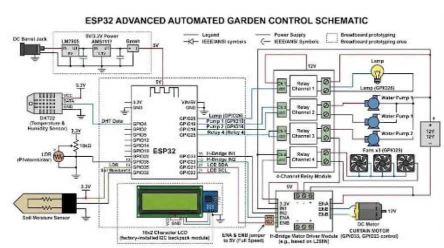


Figure 4.2: Circuit Schematic Diagram

Thermodynamically, the system prevents thermal runaway by controlling the heat rate inside the greenhouse. When the DHT22 sensor detects the temperature exceeding the set threshold, the system alters the heat accumulation **as shown in Equation 5**:

Equation 5:Net thermal accumulation model

$$q_{net} = q_{solar} - q_{conv}$$

where q_{net} is the net heat transfer rate inside the greenhouse required to represent the thermal gain or loss of the system, q_{solar} is the incoming solar heat load, which represents the energy gain from solar radiation, q_{conv} is the heat removed from the system due to the cooling fan? The actuation of the DC motor to move the shade reduces the value of q_{solar}. Additionally, the cooling fan maximizes q_{conv}, which is represented by Newton's Law of Cooling **(Equation 6)**:

Equation 6: Newton's Law of Cooling

$$q_{conv}=h.A.(T_{internal}-T_{ambient})$$

where h is the convective heat transfer coefficient established by the fan's airflow, A is the effective internal surface area, and T represents the respective internal and ambient temperature. Hydrologically, the stability of soil moisture is analyzed through the conservation of mass principle. The soil moisture sensor triggers the water pump to compensate for evaporative losses, described by the following moisture balance **equation 7**:

Equation 7: Differential mass balance equation

$$\frac{d(\theta.V)}{dt}=Q_{pump}-E_{rate}$$

where θ is the volumetric water content of the soil (m^3/m^3), V is the control volume, Q_{pump} is the volumetric flow rate provided by the pump, and E_{rate} represents the evaporative losses rate from the system (m^3/s). Furthermore, low-light conditions are corrected through an LDR sensor that activates a lamp, ensuring sufficient flux to maintain the baseline illuminance level for plant growth

The implementation of this active feedback control system is justified as it provides continuous real-time error correction compared to passive cooling or static timer-based irrigation systems, which cannot adapt effectively to dynamic environmental variables. By regulating q_{net} toward zero or negative values during peak thermal loads and adjusting the rate of change of $\frac{d(\theta.V)}{dt}$The system reduces environmental fluctuations. This closed-loop regulating mechanism helps explain the empirical results obtained, where the prototype maintained internal temperature within the target range of 15 °C to 25 °C and kept soil moisture within the optimal range of 40% to 50% under test conditions.

Conclusions:

In light of these considerations, one major contributor to environmental pollution is the limited implementation of recycling, exacerbated by increasing population density, hindering Egypt's progress. These significant issues particularly appear in the city of 6th October, resulting in elevated temperature levels of 36 °C. Thus, the proposed solution represents an even-span smart greenhouse. This design ensures good ventilation, with even temperature distribution for orchid and coriander cultivation.

Taken together, the highlighted results of three environmental parameters (soil moisture, light intensity, and temperature) include: 22.85 °C temperature, 1686 soil moisture, 57.08% humidity, and 69 LDR. All data were collected via an ESP-32-based monitoring system. The small-scale prototype was also designed more cost-effectively compared with the prior solution, such as Bluehouse, which involves high operational costs.

Recommendation

Real-Life Application:

The even-span greenhouse structure is constructed in the 6th of October City, Egypt, due to severe water scarcity and low soil quality for sustainable cultivation. **As illustrated in Figure 4.3,** the desert-like conditions are characterized by high temperature levels (approximately 36°C) and extremely low soil moisture (1%). Furthermore, the city experiences low precipitation (nearly 0.1 mm/day); meanwhile, the average is estimated at 7.5 mm/day.

Figure 4.3: 6th October City (Proposed Location)

The implementation of the **even-span** structure would emphasize air circulation and ensure uniform temperature distribution. This is primarily via the utilization of two symmetrical spans, identical slopes with equal lengths. The structure is proposed to be constructed with galvanized steel covered with fiberglass ***(ACF Greenhouses Buying Guide - Help Choosing a Greenhouse, n.d.)*** due to its durability and UV resistance. The structure is designed with dimensions of 44.0 × 7.0 m along a height of 2.0 m ***(Kwon et al., 2015).***

Advanced AI Cameras & Vision Systems

It is recommended to apply advanced AI cameras and vision systems for the detection of the condition and the appearance of diseases in the plant. These systems are supported by High-end hardware possibilities. Among these possibilities are high-resolution imaging sensors, which may extend to multispectral or hyperspectral cameras. Such sensors capture detailed spatial and spectral information beyond the visible range, enabling more accurate detection of plant stress and disease. They are also paired with sufficient processing capability to run complex algorithms locally, using special electronics, such as Raspberry Pi 4 or NVIDIA Jetson Nano. These systems depend on high-level and deep-learning models. For example, convolutional neural networks (CNNs), which require substantial computational resources, including high RAM capacity and parallel processing capabilities, to perform image analysis efficiently, unlike microcontrollers like ESP32, are designed for low-power, simple control tasks and a decrease in the level of processing power needed for real-time AI inference. This system can't be applicable in the prototype, due to the high cost ranging between 15,000 – 16,000 EGP for the low-end system, and about 50,000 – 54,000 EGP for high-end systems.

Drip Irrigation System

It is recommended to use drip irrigation systems due to their high efficiency in delivering water directly to the plant root zone in a controlled and precise manner. This method significantly improves the efficiency of water usage by reducing evaporation, surface runoff, and unnecessary water distribution, making it highly appropriate for arid environments such as 6th of October City, where water conservation is important, due to a lack of water sources. The system operates through a network of small tubes and emitters that release water drop by drop, which relies on programmed control signals from the loop circuit, which can be integrated with soil moisture, through which the sensor detects the level of the soil moisture, and then sends signals for the loop to allow the work of the system if it is required. This closed feedback mechanism helps maintain optimal soil moisture levels, enhancing plant growth stability and reducing stress caused by overwatering or underwatering. However, it is hard to apply this to the idea of the project because of the highly complex tubing infrastructure, precise clog-free maintenance, and stable, continuous water pressure.

Semiconductor LED Chips

It is recommended to use the Semiconductore LED Chips, due to their importance in providing the appropriate spectrum wavelengths according to the three phases of plant live, which are 1-seedling phase, 2-vegetative phase, 3-flowering phase, and it is a strong edit due to the enhancement leaf and stem development, improving flowering and fruiting efficiency, reducing wasted energy compared to normal white light, and allowing plants to grow in optimized, controlled environments regardless of natural sunlight. The loop circuit is preprogrammed with inputs for the plant's phases and properties, and these inputs provide a signal to the current, and finally to the semiconductor LED material, such as Gallium Nitride, to emit the desired wavelength for each phase. Finally, it can't be applied in the prototype's enclosed system, because of the high cost that ranges between 10000 and 25000 pounds for the small scale of the prototype.

Table 4.1 LOs connection

Learning Outcome	Utilization
ME.2.07	This topic shows how the Law of Conservation of Momentum explains collisions by linking force, time, and change in motion (impulse), and is used to study and predict real-life impacts like car crashes.
PH.2.09	Energy storage and control in circuits Energy storage and control in circuits RL circuits and inductors show how energy is stored in magnetic fields and released over time, which is used in devices like power supplies, electric motors, and timing circuits in electronics.
PH.2.10	Studying diodes and transistors explains how devices like chargers, computers, and smartphones control current flow, convert AC to DC, and amplify signals—forming the basis of all digital technology.
Bi.2.10	Understanding Natural Selection helps explain how organisms adapt to their environment over time. In real life, this connects to biodiversity and conservation, as studying fossils and evolution allows us to understand how species survive environmental changes and how to protect them from extinction.
CH.2.11	The studying of the hydrocarbons and cracking explains how fuels are produced and used to heat a greenhouse. This helps you choose efficient fuels, control gas levels for plant growth, and select suitable polymer materials to build a strong and effective greenhouse system.
CH.2.09	The smart greenhouse system operates using an external power supply that provides electrical energy to sensors, controllers, and actuators. Although no batteries are used, the system still depends on electric current flow from the power supply, which is based on electron movement in circuits.
PH.2.12	The smart greenhouse uses Sensors, a microcontroller (ESP32 / Arduino), a fan, a pump, an LED light, and a power supply (AC → DC adapter); all of these depend on electrical circuits and power behavior. So consequently, it is a must to analyze the prototype efficiency running system, using this LO.
PH.2.15	The system likely includes: Power regulation circuits, Protection components (diodes), Conversion from AC to DC (adapter). Which are included in this LO.
MA.2.06	There are some changes in the conditions which are sudden, like the (pump ON/OFF, fan switching), and the change can be smooth over time, like (temperature, humidity), such that changes can be studied and analyzed by the limits and continuity graphs and functions.
ME.2.06	The center of mass concept is relevant to ensuring stability of the greenhouse structure, especially when adding components like water tanks or mounted sensors, to prevent tipping or imbalance.

Portfolio Citations

1. *Atlantic Sapphire Silage Plant with BioChop System | Landia*. (n.d.). https://www.landiainc.com/case-studies/silage-plant-at-atlantic-sapphire
2. Bluehouse Salmon. (2025, April 23). *Mission - Bluehouse Salmon*. https://bluehousesalmon.com/mission/
3. *Dissolved oxygen - environmental measurement systems*. (2025, October 21). Environmental Measurement Systems. https://www.fondriest.com/environmental-measurements/parameters/water-quality/dissolved-oxygen/
4. PENETRON. (n.d.). *Penetron Dives into Salmon Farming – in Southern Florida*. https://www.penetron.com/news-media/media-releases/view/Penetron-Dives-into-Salmon-Farming-in-Southern-Florida/#:~:text=That's%20better%20than%20beef!%E2%80%9D,were%20waterproofed%20with%20PENETRON%20ADMIX.
5. Recirculating Aquaculture Salmon Network. (2025, March 28). *What is RAS? - Land-based Aquaculture for Atlantic Salmon*. Land-based Aquaculture for Atlantic Salmon -. https://salmononland.org/salmon-ras/what-is-ras/#:~:text=Fish%20produce%20ammonia%20as%20a,rapidly%20accumulate%20to%20toxic%20levels.
6. *This is how much electricity Atlantic Sapphire used per kilo of salmon*. (2021, December 23). SalmonBusiness. https://www.salmonbusiness.com/power-intensive-aquaculture-production-this-is-how-much-electricity-atlantic-sapphire-used-per-kilo-of-salmon/
7. WorldData.info. (2026, February 1). *Climate and temperatures in Egypt*.

Worlddata.info. https://www.worlddata.info/africa/egypt/climate.php

8. *6th of October City » EBRD Green Cities*. (n.d.).

https://www.ebrdgreencities.com/our cities/cities/6th-of-october-city/

9. Lisa. (2025, April 25). *How much should you plant per person? - The practical planter*. The Practical Planter.

https://thepracticalplanter.com/how-much-to-plant-per-person/

10. Mohamed, E., Ali, A., El-Shirbeny, M., Abutaleb, K., & Shaddad, S. M. (2019). Mapping soil moisture and their correlation with crop pattern using remotely sensed data in arid region. *The Egyptian Journal of Remote Sensing and Space Science*, *23*(3), 347–353. https://doi.org/10.1016/j.ejrs.2019.04.003

11. Benzzine, H., Labrim, H., Aouni, I. E., Bouali, K., Achour, Y., Saad, A., Zejli, D., & Bouayadi, R. E. (2026). Integrated Passive Cooling Techniques for Energy-Efficient Greenhouses in Hot–Arid Environments: Evidence from a Systematic Review. *Water*, *18*(4), 463. https://doi.org/10.3390/w18040463

12. *Simulated historical climate & weather data for 6th of October City – meteoblue*. (n.d.). Meteoblue.

https://www.meteoblue.com/en/weather/historyclimate/climatemodelled/6th-of-october-city_egypt_353219

13. Admin. (2025, June 18). *What is an Even-Span Greenhouse? Advantages, disadvantages, and diagram - INSONGREEN*. INSONGREEN.

https://www.insongreen.com/what- is-even-span-greenhouse/

14. Abdelzaher, M. A. (2023). Sustainable development goals for industry, innovation, and infrastructure: demolition waste incorporated with nanoplastic waste enhanced the physico-mechanical properties of white cement paste composites. Applied Nanoscience, 13(8), 5521 5536. https://doi.org/10.1007/s13204-023-02766-w

15. AfricaNews. (2023, January 18). Young Egyptians battle plastic plague. Africanews. https://www.africanews.com/2023/01/18/young egyptians-battle-plastic plague/#:~:text=The%20country%20generates%20more%20than%20three%20million%20tonnes,way%20into%20the%20Nile%20and%20the%20Mediterranean%20Sea

16. Climate change: global temperature. (2025, May 29). NOAA Climate.gov. https://www.climate.gov/news-features/understanding climate/climate-change-global-temperature

17. De Boer, M. (2023, June 5). Plastic pollution and agriculture. reNature. https://www.renature.co/articles/plastic-pollution-and-agriculture/

18. Directory, S. (2025, March 15). What are the limitations of current recycling technologies? → Question. Fashion → Sustainability Directory. https://fashion.sustainability-directory.com/question/what are-the-limitations-of-current-recycling-technologies/

19. Eco Recycling Today. (2024, October 13). How much does it cost to start a recycling business? https://www.recyclingtoday.org/blogs/news/how-much-does-it-cost to-start-a-recycling-business

20. Egypt Air Quality Index (AQI) and Air Pollution information | IQAir. (2020, July 13). https://www.iqair.com/egypt

21. *Particle Density | Soils - Part 2: Physical Properties of Soil and Soil Water - passel*. (n.d.). https://passel2.unl.edu/view/lesson/0cff7943f577/5

22. *Home | Soils - Part 2: Physical Properties of Soil and Soil Water - passel*. (n.d.). https://passel2.unl.edu/view/lesson/0cff7943f577

23. D, S. (2016, July 20). *Calculation of soil moisture (With example)*. Soil Management India. https://www.soilmanagementindia.com/soil-water/calculation-of-soil-moisture-with-example/1815

24. Modhavadia, R., Odedra, K. N., Bamaniya, J., & Jadeja, B. A. (2025). Soil-Specific Optimal sowing depth for coriander: Effects on seed

germination and early growth. *Annals of Arid Zone*, *64*(3), 413–424. https://doi.org/10.56093/aaz.v64i3.165757

25. *ACF Greenhouses Buying Guide - Help choosing a greenhouse*. (n.d.). https://www.littlegreenhouse.com/guide.shtml?utm_source=chatgpt.com

26.